Limit shapes for centrally excited walks

Ahmed Bou-Rabee, Tuan-Minh Nguyen, Yuval Peres, and Arvind Singh

ABSTRACT. A centrally excited random walk on $\mathbb{Z}^d$ moves like simple random walk, except that its first step from each site has a drift of fixed size toward the origin. Kozma (2007) conjectured that after n steps the visited set approximates a ball with radius of order $n^{1/(d+1)}$. We prove this in every dimension $d \geq 2$ and show that the rescaled numbers of visits converge uniformly to a cone, which is the potential generated by the drift on a ball. The same argument shows that a drift opposite to a subgradient of a norm produces the ball of that norm. In the plane, the outer and inner radii of the visited set differ by at most $n^{1/6}$ times a power of $\log n$, and the exponent $1/6$ cannot be lowered.

1. INTRODUCTION

Let $d \geq 2$ and $0 < \kappa < 1/d$. A **centrally excited random walk** $X_0 = 0, X_1, X_2, \ldots$ on $\mathbb{Z}^d$ moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability $1/(2d) \mp \kappa x_i/(2|x|)$ for $1 \leq i \leq d$, where $e_1, \ldots, e_d$ is the standard basis of $\mathbb{R}^d$. Given the path up to that departure, this step has mean $-\kappa x/|x|$, a drift of size κ toward the origin (Figure 1).

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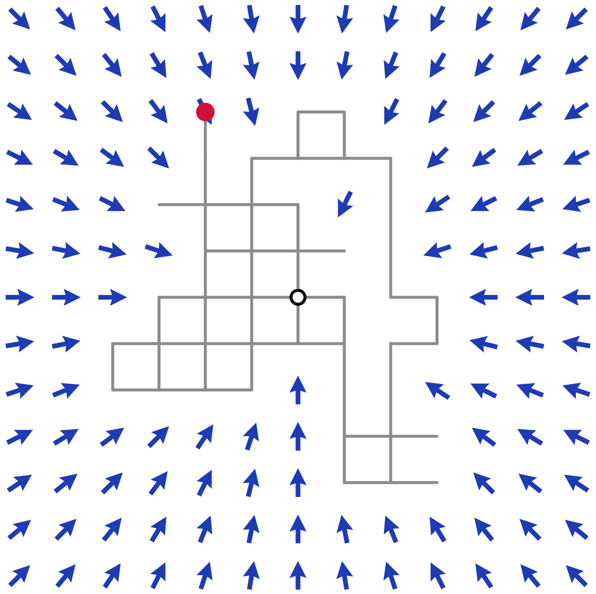

FIGURE 1. The walk after 120 steps, for $d = 2$ and $\kappa = 0.4$: its path (gray), the origin (open circle) and its position (red dot). At each site $x \neq 0$ that it has not yet left, an arrow shows the mean $-\kappa x/|x|$ of its first step from x.

At an Oberwolfach problem session, Kozma [2007] conjectured that the visited set approximates a ball of radius of order $n^{1/(d+1)}$. We prove this for every $d \geq 2$, identify the radius and the uniform limit of the local times, and bound the fluctuations of both. In the plane the inner and outer radii of the visited set fluctuate on the scale $n^{1/6}$ up to logarithmic factors; in higher dimensions their fluctuations are bounded by powers of $\log n$. More generally, a first-departure drift opposite to a subgradient of a norm produces the ball of that norm.

1.1. The limit shape.

For an integer $n \geq 0$ let

$$\ell_n(x) \coloneqq \sum_{j=0}^{n-1} \mathbf{1}_{\{X_j = x\}} \qquad \text{and} \qquad A_n \coloneqq \{x : \ell_n(x) > 0\}\,.$$

The **local time** $\ell_n(x)$ is the number of departures from x before time n, and A_n is the **range**; the set of sites visited by time n is A_{n+1}. Write $B(y, \rho)$ for the open Euclidean ball of radius ρ about y and ω_d for the volume of $B(0,1)$, and let

$$r_n \coloneqq \left(\frac{(d+1)n}{2d\kappa\omega_d}\right)^{\frac{1}{d+1}}, \tag{1.1}$$

so that the cone $2d\kappa(r_n - |v|)_+$ has integral n. Our first theorem says that the range approximates $B(0, r_n)$ and the local times approximate this cone.

Theorem 1.1 (Limit shape)**.** *Let $d \geq 2$ and $0 < \kappa < \frac{1}{d}$. Then, almost surely, the centrally excited random walk has the following properties.*

(i) Shape*: for every $0 < \eta < 1$ and all sufficiently large n,*

$$\{x \in \mathbb{Z}^d : |x| < (1-\eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : |x| < (1+\eta)r_n\}\,.$$

(ii) Local times*: as $n \to \infty$,*

$$\frac{1}{r_n} \max_{x \in \mathbb{Z}^d} \big|\ell_n(x) - 2d\kappa(r_n - |x|)_+\big| \longrightarrow 0\,.$$

(iii) Recurrence*: the walk visits every site of $\mathbb{Z}^d$ infinitely often.*

1.2. Fluctuations.

Replace each site $x \in A_n$ by the unit cell $C_x \coloneqq x + [-\frac{1}{2}, \frac{1}{2})^d$, and let $D_n \coloneqq \bigcup_{x \in A_n} C_x$. Define the **inner radius** and the **outer radius** as

$$R_{\text{in}}(n) \coloneqq \inf_{y \notin D_n} |y| \qquad \text{and} \qquad R_{\text{out}}(n) \coloneqq \max_{0 \leq j \leq n} |X_j|\,. \tag{1.2}$$

Theorem 1.2 (Fluctuation bounds)**.** *Let $d \geq 2$ and $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d, \kappa, p) < \infty$ such that, for every integer $n \geq 2$, the following bounds hold simultaneously with probability at least $1 - Cn^{-p}$.*

(i) Radii*:*

$$\begin{aligned} |R_{\text{in}}(n) - r_n| &\leq C\sqrt{r_n \log n} \quad \textit{and} \quad R_{\text{out}}(n) - r_n \leq C\sqrt{r_n}(\log n)^{5/2} \quad \textit{if } d = 2\,, \\ |R_{\text{in}}(n) - r_n| &\leq C \log n \quad \textit{and} \quad R_{\text{out}}(n) - r_n \leq C(\log n)^{d+1} \quad \textit{if } d \geq 3\,. \end{aligned}$$

(ii) Volume*:*

$$\big|r_n^{-1} D_n \triangle B(0,1)\big| \leq C \begin{cases} \sqrt{\dfrac{\log n}{r_n}}, & d = 2, \\ \dfrac{\log n}{r_n}, & d \geq 3 \end{cases}.$$

(iii) Local times*:*

$$\max_{x\in\mathbb{Z}^d}\big|\ell_n(x)-2d\kappa(r_n-|x|)_+\big|\le C\begin{cases}\sqrt{r_n}(\log n)^{3/2}, & d=2,\\ \sqrt{r_n}\log n, & d\ge 3\end{cases}.$$

In particular, these bounds hold for all sufficiently large n almost surely.

In the plane, r_n has order $n^{1/3}$, so part (i) bounds the deviations of both radii from r_n by $n^{1/6}$ times a power of $\log n$; in higher dimensions it bounds them by a power of $\log n$. The next theorem shows that the planar exponent $1/6$ cannot be lowered and that the local-time bound in part (iii) has the correct order in $d\ge 3$.

Theorem 1.3 (Sharpness)**.** *Let $d\ge 2$ and $0<\kappa<1/d$, and consider the centrally excited random walk.*

(i) Radii in the plane*: let $d=2$. For every $p>0$ there exists $c(\kappa,p)>0$ such that, for all sufficiently large n,*

$$\mathbb{P}\big(r_n-R_{\mathrm{in}}(n)\ge c\sqrt{r_n\log n}\big)\ge n^{-p}\qquad\textit{and}\qquad\mathbb{P}\big(R_{\mathrm{out}}(n)-r_n\ge c\sqrt{r_n\log n}\big)\ge n^{-p}\,. \tag{1.3}$$

(ii) Iterated logarithm for the radii in the plane*: let $d=2$. Almost surely,*

$$\limsup_{n\to\infty}\frac{r_n-R_{\mathrm{in}}(n)}{\sqrt{r_n\log\log n}}\ge\frac{1}{\sqrt{10\pi\kappa}}\qquad\textit{and}\qquad\limsup_{n\to\infty}\frac{R_{\mathrm{out}}(n)-r_n}{\sqrt{r_n\log\log n}}\ge\frac{1}{\sqrt{10\pi\kappa}}\,. \tag{1.4}$$

(iii) Local times near the origin*: there exists $c(d,\kappa)>0$ such that, for all sufficiently large n, with probability at least $1-n^{-c}$,*

$$\max_{\substack{x\in\mathbb{Z}^d\\ |x|\le r_n^{1/2}}}\big|\ell_n(x)-2d\kappa(r_n-|x|)\big|\ge c\begin{cases}\sqrt{r_n}\log n, & d=2,\\ \sqrt{r_n}\log n, & d\ge 3\end{cases}. \tag{1.5}$$

(iv) Difference of the radii in the plane*: let $d=2$.*

(a) *For every $p>0$ there exists $c(\kappa,p)>0$ such that, for all sufficiently large n,*

$$\mathbb{P}\big(R_{\mathrm{out}}(n)-R_{\mathrm{in}}(n)\ge c\sqrt{r_n\log n}\big)\ge n^{-p}\,. \tag{1.6}$$

(b) *Almost surely,*

$$\limsup_{n\to\infty}\frac{R_{\mathrm{out}}(n)-R_{\mathrm{in}}(n)}{\sqrt{r_n\log\log n}}\ge\sqrt{\frac{\pi}{3\kappa}}\,. \tag{1.7}$$

(c) *For every $a\ge 0$,*

$$\limsup_{n\to\infty}\mathbb{P}\big(R_{\mathrm{out}}(n)-R_{\mathrm{in}}(n)\le a\sqrt{r_n}\big)\le 1-\exp\Big(-\frac{3\kappa a^2}{\pi}\Big)\,. \tag{1.8}$$

Part (i) shows that the bound on $|R_{\mathrm{in}}(n)-r_n|$ in Theorem 1.2 (i) is attained, up to the constant, with probability at least n^{-p}. Part (iv) shows that the planar fluctuations are not only fluctuations of a radius: by (c), the outer and inner radii differ by more than $a\sqrt{r_n}$ with probability at least $e^{-3\kappa a^2/\pi}-o(1)$. Part (iii) matches the local-time bound of Theorem 1.2 (iii) when $d\ge 3$; in the plane we show in Section 9 that, away from the boundary, the local times deviate from the cone by at most $C\sqrt{r_n}\log n$, so part (iii) is sharp there as well.

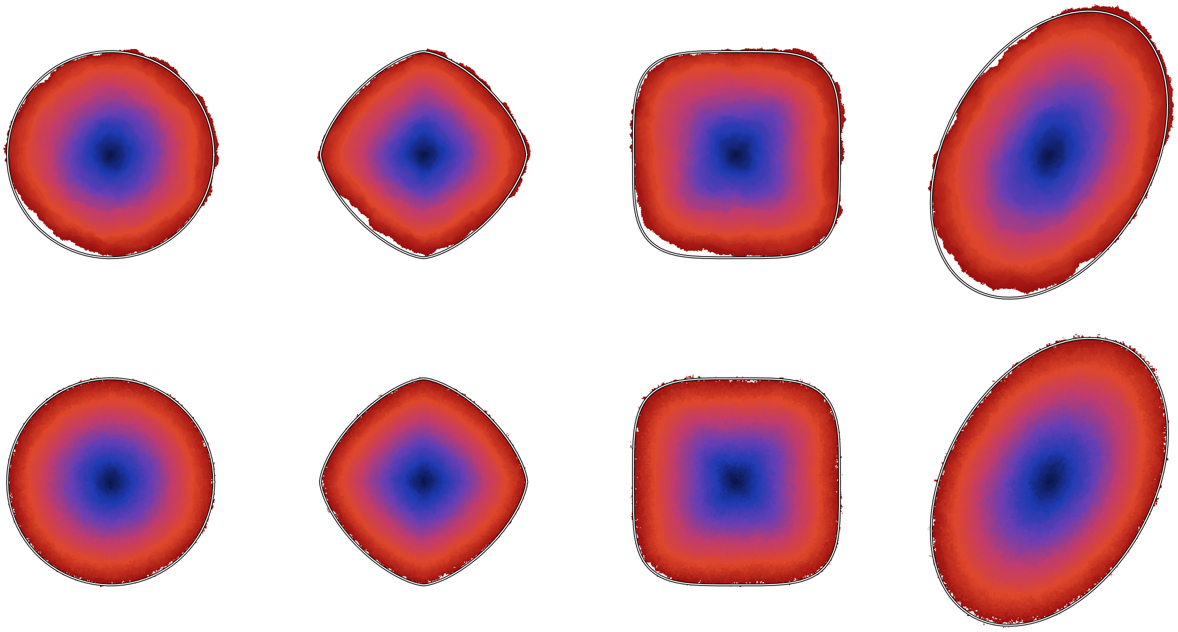

FIGURE 2. Ranges $r_n^{-1}A_n$ for the Euclidean, $\ell^{3/2}$, ℓ^4 and elliptical norms (left to right), at $n = 10^9$. Top: $d = 2$, $\kappa = 0.4$; bottom: $x_3 = 0$ sections for $d = 3$, $\kappa = 0.25$. The unit balls are outlined and the blue-to-red colors show $(t_x/n)^{1/(d+1)}$, with t_x the first-departure time.

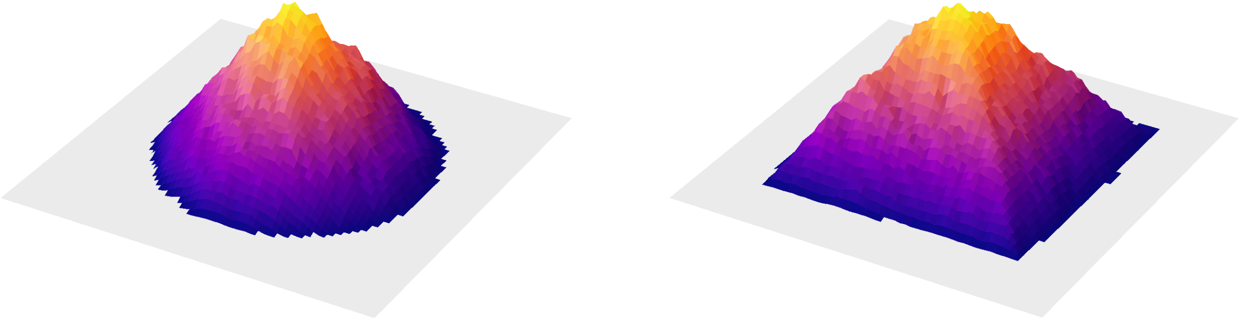

FIGURE 3. Local times $\ell_n(x)/(2d\kappa r_n)$ over x/r_n for the Euclidean norm (left) and ℓ^∞ (right): $d = 2$, $n = 10^{10}$, $\kappa = 0.4$.

1.3. **Limit shapes for other norms.** The limit shape theorem extends to drifts given by other norms. Let Ψ be a norm with unit ball B_Ψ. Replace $x/|x|$ in the first-departure probabilities by a subgradient $\xi(x)$ of Ψ at x, chosen in advance at each site, and assume $\kappa \max_i \Psi(e_i) < 1/d$. Then the range approximates $\{\Psi < r_n\}$ and the local times approximate $2d\kappa(r_n - \Psi)_+$, with $|B_\Psi|$ replacing ω_d in (1.1) (Theorem 2.1).

The theorem includes norms with corners, such as $\|\cdot\|_1$ and $\|\cdot\|_\infty$, with rates uniform in the choice of subgradients. It also extends to the Minkowski functional of every compact convex set containing the origin in its interior (end of Section 5). Figures 2 and 3 show the range and the local times.

1.4. **Strategy and key ideas.** The proof compares the local times with an explicit potential. Dynkin's formula for the Green function with pole y writes $\ell_n(y)$ as a sum over the first departures, each weighted by the inner product of its drift with the gradient of the Green function, plus a martingale and a smaller error. The continuum version of this sum is a potential: for a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$U_D(y) \coloneqq \frac{2\kappa}{\omega_d} \int_D \frac{v}{|v|} \cdot \frac{v-y}{|v-y|^d} \, \mathrm{d}v \tag{1.9}$$

be the potential generated by the vector field $-\kappa v/|v|$ on D. Integration along the rays from y shows that the potential of a ball is a cone: for every $\rho > 0$ and every $y \in \mathbb{R}^d$,

$$U_{B(0,\rho)}(y) = 2d\kappa(\rho - |y|)_+ \, . \tag{1.10}$$

Replacing $v/|v|$ by $\nabla\Psi(v)$ gives the cone $2d\kappa(\rho - \Psi(y))_+$ for a ball of the norm (Lemma 2.2).

So Dynkin's formula compares ℓ_n with U_{D_n}, up to a martingale and a smaller error. Near the boundary of the range, the martingale error bound can be larger than the local times and the potential, so this comparison alone need not locate the boundary. We use two further facts. Local times vanish outside the range, and the walk can reach a distant site only by crossing the intervening region against its first-departure drift. The main comparison is at a contact point: a point y_0 on the boundary of the largest ball about the origin contained in D_n that lies in the closure of a cell disjoint from D_n. The local time vanishes on that cell, so the potential at y_0 is small, while every point of D_n outside the ball contributes nonnegatively to it.

1. *Local times and the potential* (Section 3). Freedman's inequality controls the martingales in Dynkin's formula uniformly over sites and time intervals. Replacing the sum over first departures by U_{D_n} costs $O(\log n)$, also for norms with corners; convexity controls the variation of the subgradients across a cell (Lemma 3.3).
2. *Crossing and coarse bounds* (Section 4). Let τ_R be the time at which the walk leaves the ball of the norm with radius R. Until τ_R, and until one of the martingales in Dynkin's formula exceeds R in absolute value, Dynkin's formula bounds each local time by CR, and hence the elapsed time by CR^{d+1}; Freedman's inequality then shows that the martingales are unlikely to exceed R before τ_R. At time τ_R, the comparison at the contact point and the crossing argument show that the range contains all sites of a ball of radius cR. Integrating the cone that approximates the local times gives $\tau_R \geq cR^{d+1}$, and the coarse bounds at time n follow (Proposition 4.1).
3. *The limit shape* (Section 5). Let b be the radius of the largest ball of the norm about the origin contained in D_n. By convexity, every point v of D_n outside this ball contributes at least $c(\Psi(v) - b)r_n^{-d}$ to the potential at the contact point. Dynkin's formula at the center of the adjacent cell bounds this potential by the martingale there, and Dynkin's formula for the square of the Green function bounds the bracket of that martingale in terms of the local time at its pole, which vanishes (Lemma 5.2). So the part of D_n outside the ball has small potential, which gives the cone for the local times, and the martingale obtained from $|X_n|^2$ gives $b \approx r_n$. To show that no visited site lies far outside the ball of a norm with corners, we apply the crossing argument in the direction from the nearest point of the ball $\{\Psi \leq r_n\}$ to the visited site farthest from this ball in Euclidean distance; every subgradient at a visited site where the inner product with this direction is close to its maximum has a positive inner product with this direction.
4. *The inner radius* (Section 6). For the Euclidean norm, every point v of D_n outside the inner ball contributes at least $c|v - y_0|^{2-d}/r_n$ to the potential at the contact point y_0, a bound that does not vanish near the boundary of the ball. With Lemma 5.2, this bounds the volume of D_n outside the ball and sharpens the inner-radius estimate (Proposition 6.1).
5. *The outer radius* (Section 7). We evaluate the potential at points beyond the outer radius, where the local time vanishes. Points of D_n near the outer sphere contribute negatively to this potential. The curvature of the sphere bounds the positive contributions by an integral of $|v - y|^{2-d}/|v|$ over the part of D_n outside the inner ball, which we bound by integrating the potential along rays when $d \geq 3$ and by the volume bound of Section 6 when $d = 2$. So little of D_n lies near

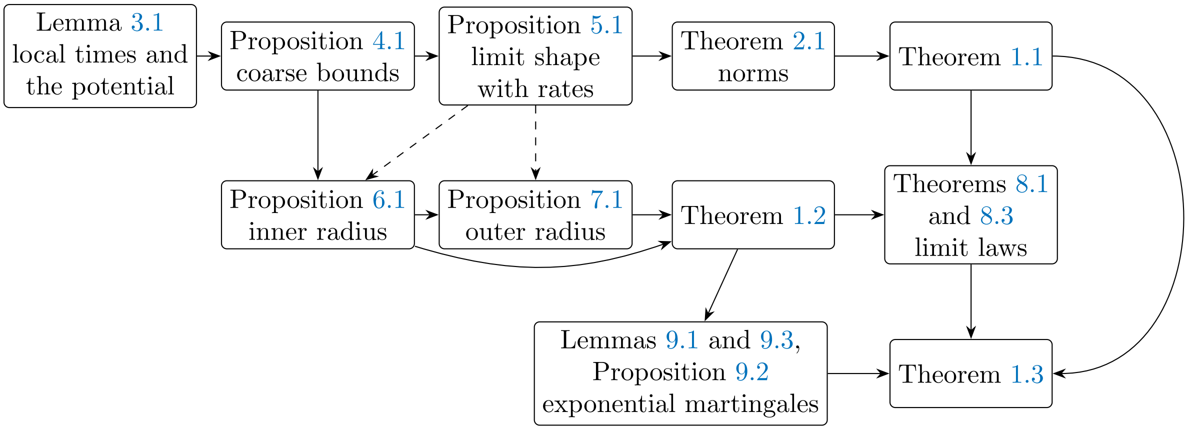

FIGURE 4. Dependencies among the main results. A solid arrow means that a result is used; a dashed arrow means that estimates from Section 5 are used.

the outer sphere. These volume bounds control the local times outside the inner ball, and the crossing argument gives the outer-radius bound (Proposition 7.1). The same estimates give the uniform local-time bound.

The cone determines the limits of the brackets of the martingales in Dynkin's formula and of the martingale obtained from $|X_n|^2$, so the martingale central limit theorem and the law of the iterated logarithm give the limit laws of Section 8. The lower bounds need deviations of about $(\log n)^{1/2}$ standard deviations, which we obtain from exponential martingales: the martingale obtained from $|X_n|^2$ controls the planar radii, and many weakly correlated martingales in Dynkin's formula control the largest deviation of the local times near the origin. The difference of the radii is controlled by the vector martingale $X_n + \kappa \sum_{x \in A_n} x/|x|$: the vectors $x/|x|$ cancel inside the inner disk, so fluctuations of order $\sqrt{n}$ of this martingale force fluctuations of order $\sqrt{r_n}$ of the difference of the radii, and its two-dimensional Gaussian limit gives the bound in Theorem 1.3 (iv)(c). Figure 4 records the dependencies.

The local times play the role of the odometer of the divisible sandpile, which Levine and Peres [2009] compared with an explicit potential. Martingales with a pole at an unvisited site, and separate estimates that rule out thin tentacles, also appear in the bounds of Jerison, Levine, and Sheffield [2012, 2013] for internal diffusion-limited aggregation. Our logarithmic lower bounds use the inner/outer deviation dichotomy of Asselah and Gaudillière [2014, Proposition 1.2].

1.5. **Related work.** Kozma [2013] reported recurrence of the walk excited to the center, citing unpublished work from 2006. Dembo, Huang, and Sidoravicius [2014a] described the manuscript of Kozma [2006] for a related walk in $d \geq 3$, with drift $-\kappa x/\|x\|_1$ at the first visit to a site having an unvisited neighbor; Dembo, Huang, and Sidoravicius [2014b] cited it for recurrence in every dimension. We take the name centrally excited random walk from the title of that manuscript. The drift of its walk has a Euclidean length that depends on the direction, whereas our drift has length κ. Benjamini, Duminil-Copin, Kozma, and Lucas [2020] also recorded the predicted radius $n^{1/(d+1)}$.

For the excited random walk with first-visit drift in a fixed direction, Benjamini and Wilson [2003] proved transience in $d \geq 2$ and positive lower speed in $d \geq 4$. Kozma [2003, 2005] obtained positive lower speed in $d = 3$ and $d = 2$, and Bérard and Ramírez [2007] proved laws of large numbers and central limit theorems. Menshikov, Popov, Ramírez, and Vachkovskaia [2012] extended positive-speed results to uniformly elliptic walks with bounded jumps and uniformly positive first-visit drift in a fixed direction and zero drift at later visits. Zerner [2005, 2006] gave recurrence/transience criteria on $\mathbb{Z}$ and strips for walks with multiple cookies. Amir, Benjamini, and Kozma [2008] proved recurrence for a walk in the upper half of $\mathbb{Z}^3$ that steps down at new sites. See Kosygina and Zerner [2013] for a survey.

The radius $n^{1/(d+1)}$ and range size $n^{d/(d+1)}$ are also predicted for once-reinforced walks with strong reinforcement [Beffara, 2011; Elboim and Kozma, 2026], and supported by simulations [Foster, Grassberger, and Paczuski, 2009]; related predictions for self-attracting walks appear in Ordemann, Berkolaiko, Havlin, and Bunde [2000]; Ordemann, Tomer, Berkolaiko, Havlin, and Bunde [2001]. For once-reinforced random walk on $\mathbb{Z}^d$ with $d \geq 2$, Bou-Rabee and Peres [2026c] proved the lower bound $cn^{d/(d+1)}$ on the expected range for every reinforcement parameter. The matching upper bound and limit shape remain conjectural. See Pemantle [2007] for reinforced processes.

Dembo, Groisman, Huang, and Sidoravicius [2021] proved an averaging principle and shape theorem for growth models motivated by origin-excited and once-reinforced walks. Their simulations with inward first steps along a coordinate of largest absolute value or along all coordinates produced squares and diamonds. Our norm theorem proves these shapes for first-departure drifts opposite to subgradients of $\|\cdot\|_\infty$ and $\|\cdot\|_1$. For related continuum growth models, Dembo and Yang [2025a,b] obtained limits described by geometric flow and regularized Kardar–Parisi–Zhang equations.

Internal diffusion-limited aggregation [Lawler, Bramson, and Griffeath, 1992] and rotor-router aggregation [Levine and Peres, 2009] have Euclidean ball limit shapes. The former has radius fluctuations of order at most $\log r$ in the plane and $(\log r)^{1/2}$ in higher dimensions [Jerison et al., 2012, 2013; Asselah and Gaudillière, 2013a,b], and its rescaled fluctuations approach a variant of the Gaussian free field [Jerison, Levine, and Sheffield, 2014]. For the centrally excited walk, Theorem 1.3 gives much larger planar fluctuations of the radii and of their difference; the fixed-site covariance in Theorem 8.3 is again expressed through the Green function. See Levine and Peres [2017] for aggregation models.

For Eulerian walkers [Priezzhev, Dhar, Dhar, and Krishnamurthy, 1996], simulations suggested a circular planar range of radius of order $n^{1/3}$ after n steps [Kapri and Dhar, 2009]. Florescu, Levine, and Peres [2016] proved that every simple rotor walk on $\mathbb{Z}^d$ visits at least $cn^{d/(d+1)}$ sites in n steps. For clockwise planar rotors with independent uniform initial states, Bou-Rabee and Peres [2026a] proved convergence of the rescaled range to a deterministic compact convex set; whether it is a disk remains open.

1.6. **Notation and conventions.**

- The Euclidean norm is $|x|$, and $\|x\|_p$ is the ℓ^p norm for $1 \leq p \leq \infty$. The cardinality of a set $A \subset \mathbb{Z}^d$ is $|A|$, and the Lebesgue measure of a set $D \subset \mathbb{R}^d$ is $|D|$. The symmetric difference is $\triangle$, and $t_+ := \max\{t, 0\}$ and $s \wedge t := \min\{s, t\}$.
- Let $u_x := x/|x|$ for $x \in \mathbb{R}^d \setminus \{0\}$ and $u_0 := 0$, and let $\xi(0) := 0$.
- The letters $c > 0$ and $C > 0$ denote constants depending only on d, κ and the exponent p of the failure probability under discussion, and in Sections 2–5 also on the norm Ψ, but not on the choice of the subgradients $\xi(x)$; they may change from line to line. Every further dependence is stated, and a subscript on O, as in O_y, means that the implied constant also depends on y. We write $a_n \asymp b_n$ if $c \leq a_n/b_n \leq C$.
- The σ-field generated by $X_0, \ldots, X_j$ is $\mathcal{F}_j$. For martingales Z_1 and Z_2, the brackets $\langle Z_1 \rangle_n$ and $\langle Z_1, Z_2 \rangle_n$ are the predictable quadratic variation and covariation through time n.
- The indicator that the step at time j is a first departure is $I_j := \mathbf{1}_{\{X_j \notin A_j\}}$, so that $\mathbb{E}(X_{j+1} - X_j \mid \mathcal{F}_j) = -\kappa I_j \xi(X_j)$, where $\xi(x) = u_x$ for the Euclidean norm.
- The function $\widetilde{\ell}_n$ on $\mathbb{R}^d$ equals $\ell_n(x)$ on C_x for every $x \in \mathbb{Z}^d$; it vanishes off D_n.
- All estimates concern integers $n \geq 2$, so that $\log n > 0$. In a statement that holds with probability at least $1 - Cn^{-p}$ for all sufficiently large n, enlarging C makes it hold for every $n \geq 2$.

1.7. **Lean formalization.** Every labeled theorem, lemma, proposition and corollary of this paper has been formalized in Lean 4. The formalization builds on Mathlib. No proof uses `sorry` or an axiom beyond the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The development is at `https://github.com/nitromannitol/Centrally-Excited-Random-Walk`.

Use of artificial intelligence. The authors developed the main ideas of this work and directed its mathematical development. Throughout the project, the authors used large language models as research assistants: to explore specific questions arising in the proofs, to check the arguments and the cited results, and to audit drafts of the paper. The exposition, the simulations and the figures were produced with AI agents, primarily Claude (Anthropic), under the close supervision of the authors. The authors take responsibility for the mathematical content and presentation of the paper.

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Earlier versions and related models. This paper develops the results of Bou-Rabee and Peres [2026b], with simpler proofs using the stopping argument of Singh [2026]. Singh allowed more general first-departure laws with the same radial mean. Nguyen [2026] considered a different excitation rule and obtained an ℓ^1 ball as the limit shape.

2. Drifts given by a norm

In this section we give the precise setting for a drift opposite to a subgradient of a norm and state the limit shape theorem in this setting, Theorem 2.1, of which Theorem 1.1 is the Euclidean case. We also prove the identity for the potential of a ball of the norm on which its proof rests (Lemma 2.2). Sections 3–5 work in this setting and prove Theorem 2.1 at the end of Section 5.

Let Ψ be a norm on $\mathbb{R}^d$ and let B_Ψ be its unit ball. A **subgradient** of Ψ at $x \in \mathbb{R}^d$ is a vector $\xi \in \mathbb{R}^d$ such that $\Psi(y) \geq \Psi(x) + \xi \cdot (y - x)$ for every $y \in \mathbb{R}^d$, and $\partial\Psi(x)$ is the set of subgradients at x; if Ψ is differentiable at x, then $\partial\Psi(x) = \{\nabla\Psi(x)\}$. At each site $x \neq 0$ fix $\xi(x) \in \partial\Psi(x)$, with coordinates $\xi_i(x)$. The **centrally excited random walk with norm** Ψ moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability

$$\frac{1}{2d} \mp \frac{\kappa}{2}\xi_i(x) \qquad \text{for } 1 \leq i \leq d\,. \tag{2.1}$$

We assume that

$$\kappa \max_{1 \leq i \leq d} \Psi(e_i) < \frac{1}{d}\,. \tag{2.2}$$

Every $\xi \in \partial\Psi(x)$ satisfies Euler's relation $\xi \cdot x = \Psi(x)$, so the definition of a subgradient gives

$$\xi \cdot y \leq \Psi(y) \qquad \text{for every } y \in \mathbb{R}^d\,, \tag{2.3}$$

with equality at $y = x$. Hence $|\xi_i(x)| \leq \Psi(e_i)$, and (2.2) makes the probabilities (2.1) positive. The step at the first departure from x has conditional mean $-\kappa\xi(x)$, whose inner product with x is $-\kappa\Psi(x) < 0$. The Euclidean norm, with $\xi(x) = x/|x|$, gives the walk of Section 1, and then (2.2) reads $\kappa < {}^1\!/d$. Let

$$r_n := \left(\frac{(d+1)n}{2d\kappa|B_\Psi|}\right)^{1/(d+1)}\,, \tag{2.4}$$

which is (1.1) when Ψ is the Euclidean norm; the cone $2d\kappa(r_n - \Psi(v))_+$ has integral n over $\mathbb{R}^d$.

In Sections 2–5, Ψ is a norm, $\kappa > 0$ satisfies (2.2), r_n is given by (2.4), and U_D is the potential (2.5) defined below; at a point $v \in \mathbb{R}^d$, $\nabla\Psi(v)$ is the gradient, which exists for almost every v. Let $\Lambda_\Psi := \max_{|u|=1} \Psi(u)$ and $c_\Psi := \min_{|u|=1} \Psi(u)$; by (2.3), every subgradient ξ of Ψ satisfies $|\xi| \leq \Lambda_\Psi$.

In Sections 6–9, Ψ is the Euclidean norm, so that $\xi(x) = u_x$, r_n is given by (1.1), and U_D is the potential (1.9).

Theorem 2.1 (Limit shape for a norm)**.** *Let $d \geq 2$, let Ψ be a norm on $\mathbb{R}^d$, let $\xi(x) \in \partial\Psi(x)$ be chosen at the sites $x \in \mathbb{Z}^d \setminus \{0\}$, let $\kappa > 0$ satisfy (2.2), and let r_n be as in (2.4). Then, almost surely, the centrally excited random walk with norm Ψ has the following properties.*

(i) Shape*: for every $0 < \eta < 1$ and all sufficiently large n,*

$$\{x \in \mathbb{Z}^d : \Psi(x) < (1-\eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : \Psi(x) < (1+\eta)r_n\}\,.$$

(ii) Local times*: as $n \to \infty$,*

$$\frac{1}{r_n}\max_{x\in\mathbb{Z}^d}\bigl|\ell_n(x) - 2d\kappa(r_n - \Psi(x))_+\bigr| \longrightarrow 0\,.$$

(iii) Recurrence*: the walk visits every site of $\mathbb{Z}^d$ infinitely often.*

Proposition 5.1 gives rates: for every $p > 0$ there is a constant $C(d,\kappa,\Psi,p)$, not depending on the choice of ξ, such that with probability at least $1 - Cn^{-p}$ the maximum in part (ii) is at most $Cr_n^{5/6}(\log n)^{1/6}$ if $d = 2$ and $Cr_n^{d/(d+1)}(\log n)^{1/(d+1)}$ if $d \geq 3$, and the inclusions of part (i) hold with ηr_n replaced by $\log n$ times this bound on the maximum. Part (i) gives $|A_n|/r_n^d \to |B_\Psi|$, and for the Euclidean norm Theorem 1.2 gives smaller error bounds.

2.1. **The potential of a norm ball.** For a norm Ψ, the potential (1.9) is defined with the gradient of Ψ in place of $v/|v|$: for a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$U_D(y) := \frac{2\kappa}{\omega_d}\int_D \nabla\Psi(v)\cdot\frac{v-y}{|v-y|^d}\,\mathrm{d}v \tag{2.5}$$

be the potential generated by the vector field $-\kappa\nabla\Psi$ on D.

For every Lipschitz function f with compact support and every $y \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d}\nabla f(v)\cdot\frac{v-y}{|v-y|^d}\,\mathrm{d}v = -d\omega_d f(y)\,. \tag{2.6}$$

To prove it, write $v = y + tu$ with $t > 0$ and $|u| = 1$, so that $(v-y)|v-y|^{-d}\,\mathrm{d}v = u\,\mathrm{d}t\,\mathrm{d}S(u)$, where $\mathrm{d}S$ is surface measure on the unit sphere. By Rademacher's and Fubini's theorems, for almost every u the Lipschitz function $t \mapsto f(y+tu)$ has derivative $\nabla f(y+tu)\cdot u$ for almost every $t > 0$; this function vanishes for large t, so its derivative integrates to $-f(y)$ over $t > 0$, and the unit sphere has area $d\omega_d$.

Lemma 2.2 (The potential of a norm ball)**.** *Let Ψ be a norm, let $\rho > 0$, and let $y \in \mathbb{R}^d$. Then*

$$U_{\{\Psi<\rho\}}(y) = 2d\kappa(\rho - \Psi(y))_+\,. \tag{2.7}$$

Proof. The set $\{\Psi < \rho\}$ is bounded, and $\{\Psi = \rho\}$ has measure zero because Ψ is homogeneous, so $\nabla(\rho-\Psi)_+ = -\mathbf{1}_{\{\Psi<\rho\}}\nabla\Psi$ almost everywhere. Applying (2.6) to the Lipschitz function $f = (\rho-\Psi)_+$, which has compact support, proves the lemma. ☐

For the Euclidean norm, Lemma 2.2 is the identity (1.10).

3. Local times and the potential

In this section we prove Lemma 3.1, which bounds the local times on every time interval by the number of first departures in it and compares the local times with the potential U_{D_n}. Dynkin's formula expresses each local time as U_{D_n} minus a martingale, up to an error term. We bound the bracket of this martingale by a weighted sum of the local times, and use convexity to bound the error from replacing the subgradients $\xi(x)$ chosen at the sites by the gradient $\nabla\Psi$ on the cells (Lemma 3.3).

For integers $0 \le s < t \le n$, let

$$M_{s,t} := \max_{x\in\mathbb{Z}^d} \sum_{s\le j<t} \mathbf{1}_{\{X_j=x\}} \qquad \text{and} \qquad k_{s,t} := \sum_{s\le j<t} I_j\,.$$

Lemma 3.1. *Let $p > 0$. There exists $C(d,\kappa,\Psi,p) < \infty$ such that, for every integer $n \ge 2$, the following estimates hold simultaneously with probability at least $1 - Cn^{-p}$.*

(i) Largest local time*:*

$$\max_{x\in\mathbb{Z}^d} \ell_n(x) \le C|A_n|^{1/d} + C(\log n)^2\,. \tag{3.1}$$

(ii) Local times on time intervals*: for all integers $0 \le s < t \le n$,*

$$M_{s,t} \le Ck_{s,t}^{1/d} + C(\log n)^2\,. \tag{3.2}$$

(iii) Approximation by the potential*: for every $y \in \mathbb{R}^d$ with $|y| \le 2n$,*

$$|\widetilde{\ell}_n(y) - U_{D_n}(y)| \le C\log n + C \begin{cases} \left(\max_x \ell_n(x)\right)^{1/2}\log n, & d = 2,\\ \left(\max_x \ell_n(x)\log n\right)^{1/2}, & d \ge 3 \end{cases}. \tag{3.3}$$

Once $\max_x \ell_n(x) \le Cr_n$, as on the event of Proposition 4.1, the right side of (3.3) is at most $C\sqrt{r_n}\log n = o(r_n)$.

3.1. **Dynkin's formula for the local times.** Let $\Delta f(x) := (2d)^{-1}\sum_{|e|=1}(f(x+e) - f(x))$, summed over the unit vectors $e \in \mathbb{Z}^d$, be the discrete Laplacian. For $d = 2$, let g be the potential kernel of simple random walk, normalized by $g(0) = 0$. For $d \ge 3$, let $g := -G$, where the Green function $G(x)$ is the expected number of visits to x by simple random walk from the origin. In both cases $\Delta g = \mathbf{1}_{\{0\}}$; for $d = 2$ see Lawler and Limic [2010, Proposition 4.4.2]. By Lawler and Limic [2010, Theorems 4.3.1 and 4.4.4], as $|x| \to \infty$,

$$g(x) = \begin{cases} \dfrac{2}{\pi}\log|x| + c_g + O(|x|^{-2}), & d = 2,\\ -\dfrac{2}{(d-2)\omega_d}|x|^{2-d} + O(|x|^{-d}), & d \ge 3 \end{cases}, \tag{3.4}$$

where c_g is a constant. Let $\bar\nabla g(x)$ be the vector with coordinates $(g(x+e_i) - g(x-e_i))/2$ for $1 \le i \le d$. Then

$$\bar\nabla g(x) = \frac{2}{\omega_d}\frac{x}{|x|^d} + O(|x|^{-d}) \quad \text{and} \quad |g(x+e) - g(x)| \le C(1+|x|)^{1-d}\,, \tag{3.5}$$

the first for $x \ne 0$, the second for every $x \in \mathbb{Z}^d$ and every unit vector $e \in \mathbb{Z}^d$; see Lawler and Limic [2010, Corollaries 4.3.3 and 4.4.5].

For $f\colon \mathbb{Z}^d \to \mathbb{R}$ with bounded nearest-neighbor increments, let

$$\mathcal{M}_t^f := f(X_t) - f(X_0) - \sum_{j<t} \mathbb{E}\big(f(X_{j+1}) - f(X_j) \bigm| \mathcal{F}_j\big)$$

be the martingale in Dynkin's formula for f. By (2.1), the conditional expectation in the sum is $\Delta f(X_j) - \kappa I_j \xi(X_j)\cdot\bar\nabla f(X_j)$, with $\bar\nabla f$ defined like $\bar\nabla g$. For $y \in \mathbb{Z}^d$ write $g_y := g(\cdot - y)$ and $\mathcal{M}^y := \mathcal{M}^{g_y}$. Since $\Delta g_y = \mathbf{1}_{\{y\}}$ and each site of A_n has its first departure before time n,

$$\ell_n(y) = g(X_n - y) - g(-y) + \kappa\sum_{x\in A_n} \xi(x)\cdot\bar\nabla g(x-y) - \mathcal{M}_n^y\,. \tag{3.6}$$

The difference $g(X_n - y) - g(-y)$ is $O(\log n)$ for $d = 2$ and $O(1)$ for $d \geq 3$, uniformly over $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. By (3.5), the increments of $\mathcal{M}^y$ are bounded, and a departure from x adds at most $C(1 + |x - y|)^{2-2d}$ to the bracket of $\mathcal{M}^y$; so, for all integers $0 \leq s < t$,

$$\langle \mathcal{M}^y \rangle_t - \langle \mathcal{M}^y \rangle_s \leq C \sum_{s \leq j < t} (1 + |X_j - y|)^{2-2d} . \tag{3.7}$$

Lemma 3.2. *Let $p > 0$. There exists $C(p) < \infty$ such that the following holds for every integer $n \geq 2$ and every real martingale Z whose increments have absolute value at most 1: with probability at least $1 - Cn^{-p}$,*

$$|Z_n - Z_0| \leq C\big(\sqrt{\langle Z \rangle_n \log n} + \log n\big) . \tag{3.8}$$

Proof. For $0 \leq i \leq \log_2 n + 1$, Freedman [1975, Theorem 1.6] bounds the probability that $\langle Z \rangle_n \leq 2^i$ and $|Z_n - Z_0| > C(p)(\sqrt{2^i \log n} + \log n)$ by $2n^{-p-1}$. Since $\langle Z \rangle_n \leq n$, a union bound over these $O(\log n)$ values of i proves (3.8). ☐

We apply Lemma 3.2 to constant multiples of possibly stopped martingales, with union bounds over polynomially many deterministic sites and time intervals; the resulting estimates then hold also at sites and intervals that depend on the path.

Let μ be the distributional Laplacian of Ψ on $\mathbb{R}^d$, so that $\int \phi \, \mathrm{d}\mu = \int \Psi(v) \sum_{i=1}^d \partial_i^2 \phi(v) \, \mathrm{d}v$ for every smooth compactly supported function ϕ on $\mathbb{R}^d$. It is a positive Radon measure, because the smooth convex functions obtained by mollifying Ψ have nonnegative Laplacian. For $y \in \mathbb{R}^d$ and $R > 0$, let ϕ be smooth with $0 \leq \phi \leq 1$, $\phi = 1$ on $B(y, R)$, support in $B(y, 2R)$ and $|\nabla \phi| \leq 2/R$. Integrating by parts and using $|\nabla \Psi| \leq \Lambda_\Psi$ gives

$$\mu(B(y, R)) \leq \int \phi \, \mathrm{d}\mu = -\int \nabla \Psi \cdot \nabla \phi \leq 2^{d+1} \omega_d \Lambda_\Psi R^{d-1} . \tag{3.9}$$

Lemma 3.3. *There exists $C(d) < \infty$ such that, for every $x \in \mathbb{Z}^d$,*

$$\int_{C_x} |\nabla \Psi(v) - \xi(x)| \, \mathrm{d}v \leq C \mu\big(B(x, 6\sqrt{d})\big) .$$

Proof. The vector $\xi(x)$ is a subgradient of Ψ at x; for $x = 0$, where $\xi(0) = 0$, this holds because $\Psi \geq 0$. So $h(v) \coloneqq \Psi(v) - \Psi(x) - \xi(x) \cdot (v - x)$ is convex and nonnegative, $h(x) = 0$, $\nabla h = \nabla \Psi - \xi(x)$ almost everywhere, and the distributional Laplacian of h is μ. At every point v where h is differentiable, convexity of $s \mapsto h(v + se_i)$ and $h \geq 0$ give $|\partial_i h(v)| \leq h(v + e_i) + h(v - e_i)$. The cells $C_{x \pm e_i}$ are disjoint and lie in $B(x, A)$, where $A \coloneqq 2\sqrt{d}$, so integrating over C_x and summing over i gives

$$\int_{C_x} |\nabla h(v)| \, \mathrm{d}v \leq \int_{B(x,A)} h(v) \, \mathrm{d}v .$$

Let h_r be the convolution of h with a smooth mollifier supported in $B(0, r)$, where $0 < r < A$. For a unit vector u, the derivative of the convex function $s \mapsto h_r(x + su)$ is nondecreasing, so $h_r(x + su) \leq h_r(x) + s\big(u \cdot \nabla h_r(x + 2Au)\big)$ for $0 \leq s \leq A$. The Laplacian of h_r is the convolution of μ with the mollifier, so by the divergence theorem the average of $u \cdot \nabla h_r(x + 2Au)$ over unit vectors u is at most $C\mu(B(x, 2A + r))$. Integrating in polar coordinates gives $\int_{B(x,A)} h_r \leq \omega_d A^d h_r(x) + C\mu(B(x, 3A))$. Letting $r \to 0$, and using that $h_r \to h$ locally uniformly and $h(x) = 0$, proves the lemma. ☐

Proof of Lemma 3.1. Step 1: The martingales on time intervals. Let $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. Every site visited before time n lies within distance $4n$ of y, and the weights $(1 + |x - y|)^{2-2d}$ of the sites x with $|x - y| \leq 4n$ sum to $O(\log n)$ for $d = 2$ and $O(1)$ for $d \geq 3$. So (3.7) bounds $\langle \mathcal{M}^y \rangle_t - \langle \mathcal{M}^y \rangle_s$ by $CM_{s,t}$ times the sum of these weights. Lemma 3.2 for the martingales $j \mapsto \mathcal{M}^y_{(s+j) \wedge t} - \mathcal{M}^y_s$, with a union bound over the $O(n^{d+2})$ pairs of an interval and a site y, gives an event of probability at least $1 - Cn^{-p}$ on which, for all integers $0 \le s < t \le n$ and all $y \in \mathbb{Z}^d$ with $|y| \le 3n$,

$$|\mathcal{M}^y_t - \mathcal{M}^y_s| \le C \log n + C \begin{cases} \sqrt{M_{s,t}} \log n, & d = 2, \\ \sqrt{M_{s,t} \log n}, & d \ge 3 \end{cases} . \tag{3.10}$$

We work on this event.

Step 2: Local times on time intervals. For every $y \in \mathbb{R}^d$ and every set of k lattice sites, the sum of $(1 + |x-y|)^{1-d}$ over the set is at most $Ck^{1/d}$, since it is largest for the k sites nearest to y. Subtract (3.6) with s in place of n from (3.6) with t in place of n, so that the left side becomes the number of departures from y at the times $s \le j < t$, and maximize over the sites y with $|y| \le 3n$, which include all sites visited before time n. The differences of the terms $g(X_t - y)$ and $g(X_s - y)$ contribute $O(\log n)$, and (3.10) bounds the martingale terms. The difference of the sums over A_t and A_s runs over the $k_{s,t}$ sites of $A_t \setminus A_s$, so by the bound above and (3.5) it is at most $Ck_{s,t}^{1/d}$. Young's inequality absorbs the term with $\sqrt{M_{s,t}}$, which proves (3.2). Taking $s = 0$ and $t = n$ gives (3.1), since $M_{0,n} = \max_x \ell_n(x)$ and $k_{0,n} = |A_n|$.

Step 3: Comparison with the potential. We first show that every $y \in \mathbb{Z}^d$ with $|y| \le 3n$ satisfies, deterministically,

$$\ell_n(y) = U_{D_n}(y) - \mathcal{M}^y_n + O(\log n) . \tag{3.11}$$

By (3.6) and the bound on $g(X_n - y) - g(-y)$, it suffices to turn the sum $\kappa \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} g(x-y)$ into $U_{D_n}(y)$ by three replacements, each of which changes the sum by $O(\log n)$.

The summand at $x = y$ vanishes because $\bar{\nabla} g(0) = 0$. For $x \ne y$, first replace $\bar{\nabla} g(x-y)$ by $({}^{2}/_{\omega_d})(x-y)/|x-y|^d$; by (3.5), the errors sum to at most $C \sum_{|x| \le n} (1 + |x-y|)^{-d} \le C \log n$. Second, replace each kernel value by its average over the cell C_x: for cells at distance at least one from y each error is $O((1+|x-y|)^{-d})$, and the cells near y, including C_y if $y \in A_n$, contribute $O(1)$ because $|v-y|^{1-d}$ is locally integrable.

The third replacement, of $\xi(x)$ by $\nabla \Psi(v)$ for $v \in C_x$, changes the sum by at most a constant times

$$\sum_{x \in A_n} \int_{C_x} |v-y|^{1-d} |\xi(x) - \nabla \Psi(v)| \, \mathrm{d}v .$$

The terms with $|x-y| \le 8\sqrt{d}$ contribute $O(1)$, because $|\xi(x)|$ and $|\nabla \Psi(v)|$ are at most Λ_Ψ and $|v-y|^{1-d}$ is locally integrable. If $|x-y| > 8\sqrt{d}$, then $|v-y| \ge |x-y|/2$ for $v \in C_x$, and $|x-y|/4 \le |z-y| \le 2|x-y|$ for $z \in B(x, 6\sqrt{d})$; so Lemma 3.3 bounds the summand of x in the last display by $C \int_{B(x,6\sqrt{d})} |z-y|^{1-d} \, \mathrm{d}\mu(z)$. Each point of $\mathbb{R}^d$ lies in at most C of the balls $B(x, 6\sqrt{d})$ with $x \in \mathbb{Z}^d$, and those with $|x| \le n$ and $|x-y| > 8\sqrt{d}$ lie in $\{z : 1 \le |z-y| \le 8n\}$. So the terms with $|x-y| > 8\sqrt{d}$ contribute at most $C \int_{\{1 \le |z-y| \le 8n\}} |z-y|^{1-d} \, \mathrm{d}\mu(z) \le C \log n$, because (3.9) bounds the integral over each of the $O(\log n)$ annuli $\{2^k \le |z-y| < 2^{k+1}\}$ by a constant. After the three replacements the sum has become $U_{D_n}(y)$, which proves (3.11).

Splitting the integrals defining $U_{D_n}(y)$ and $U_{D_n}(z)$ at distance $2\sqrt{d}$ from y and z, as for the cell averages, gives, for all $y, z \in \mathbb{R}^d$ with $|y-z| \le \sqrt{d}$ and $|y|, |z| \le 3n$,

$$|U_{D_n}(y) - U_{D_n}(z)| \le C \log n . \tag{3.12}$$

Now let $y \in \mathbb{R}^d$ with $|y| \le 2n$, and let z be the site with $y \in C_z$, so that $\widetilde{\ell}_n(y) = \ell_n(z)$. Then (3.11) at z, (3.10) with $s = 0$ and $t = n$, and (3.12) prove (3.3). $\square$

The identity (3.11) and the bound (3.12) are deterministic. For every $p > 0$, Lemma 3.2, applied to constant multiples of the martingales $\mathcal{M}^y$, gives, with probability at least $1 - Cn^{-p}$, simultaneously for all $y \in \mathbb{Z}^d$ with $|y| \le 3n$,

$$|\mathcal{M}^y_n| \le C\big(\sqrt{\langle \mathcal{M}^y \rangle_n \log n} + \log n\big)\,. \tag{3.13}$$

3.2. **Bounds on the potential.** The potential of a set is bounded by its volume, and by the width of a shell of Ψ that contains it. Let $D \subset \mathbb{R}^d$ be a bounded measurable set. For every $y \in \mathbb{R}^d$, the integral of $|v-y|^{1-d}$ over D is at most its integral over the ball of volume $|D|$ centered at y, which is $C|D|^{1/d}$; since $|\nabla\Psi| \le \Lambda_\Psi$, there exists $C(d,\Psi) < \infty$ such that

$$\sup_{y\in\mathbb{R}^d} |U_D(y)| \le C\kappa |D|^{1/d}\,. \tag{3.14}$$

The function U_D is continuous, because it is the convolution of the bounded, compactly supported vector field $\mathbf{1}_D\nabla\Psi$ with a kernel that is the sum of an integrable function and a bounded one.

Convexity of Ψ along lines bounds the potential of a set that lies between two balls of Ψ about the origin.

Lemma 3.4. *Let $b \ge 0$ and $a > 0$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \le \Psi(v) \le b+a\}$. Then*

$$\sup_{y\in\mathbb{R}^d} |U_D(y)| \le 2d\kappa a\,.$$

Proof. Fix $y \in \mathbb{R}^d$, write $v = y + tu$ with $t > 0$ and $|u| = 1$, and let $h_u(t) \coloneqq \Psi(y+tu)$. As in the proof of (2.6), for almost every u the function h_u has derivative $\nabla\Psi(y+tu)\cdot u$ for almost every $t > 0$, so

$$U_D(y) = \frac{2\kappa}{\omega_d}\int_{\{|u|=1\}}\int_0^\infty \mathbf{1}_D(y+tu)h_u'(t)\,\mathrm{d}t\,\mathrm{d}S(u)\,.$$

Since h_u is convex, there is $t_0 \in [0,\infty]$ such that h_u is nonincreasing on $[0,t_0]$ and nondecreasing on $[t_0,\infty)$. On each of these intervals, the change of variables $s = h_u(t)$ bounds the integral of $\mathbf{1}_{[b,b+a]}(h_u(t))|h_u'(t)|$ by a. Since $0 \le \mathbf{1}_D(y+tu) \le \mathbf{1}_{[b,b+a]}(h_u(t))$, and $h_u' \le 0$ on the first interval and $h_u' \ge 0$ on the second, the inner integral lies in $[-a,a]$. Since the unit sphere has area $d\omega_d$, it follows that $|U_D(y)| \le (2\kappa/\omega_d)\,d\omega_d\,a = 2d\kappa a$, which proves the lemma; by Lemma 2.2, equality holds at $y = 0$ when $D = \{b \le \Psi < b+a\}$. ☐

4. Coarse bounds on the range and the local times

We prove that, with high probability, $|A_n|$ has order r_n^d, and $R_{\mathrm{out}}(n) = \max_{j\le n}|X_j|$ and the largest local time have order r_n (Proposition 4.1). Fix $R \ge 2$. Until the walk reaches $\{\Psi \ge R\}$, and until one of the martingales $\mathcal{M}^y$ in (3.6) reaches R in absolute value, Dynkin's formula (3.6) bounds every local time by CR, and hence the elapsed time by CR^{d+1}; Freedman's inequality then shows that the martingales are unlikely to reach R first. When the walk reaches $\{\Psi \ge R\}$, the potential at the contact point (Section 4.2) bounds the local times at the sites outside the largest ball of Ψ about the origin contained in the cells of the range. The walk has reached $\{\Psi \ge R\}$ by crossing these sites against the drift, so the crossing estimate of Section 4.1 shows that the radius of this ball has order R. Integrating the cone that approximates the local times shows that the elapsed time is at least cR^{d+1}, and the coarse bounds at time n follow by taking for R a large multiple of r_n.

Proposition 4.1. *Let $p > 0$. There exist $c(d,\kappa,\Psi,p) > 0$ and $C(d,\kappa,\Psi,p) < \infty$ such that, for every integer $n \ge 2$, with probability at least $1 - Cn^{-p}$,*

$$cr_n^d \le |A_n| \le Cr_n^d, \qquad cr_n \le \max_{x\in\mathbb{Z}^d}\ell_n(x) \le Cr_n, \qquad cr_n \le R_{\mathrm{out}}(n) \le Cr_n\,.$$

4.1. **Crossing a region with few visited sites.** A crossing against the drift requires a large increment of the martingale

$$Z_t := X_t + \kappa \sum_{j<t} I_j \xi(X_j)\,, \tag{4.1}$$

which takes values in $\mathbb{R}^d$ and has bounded increments. Azuma's inequality in each coordinate and a union bound over the pairs $s < t$ give, for every $p > 0$, with probability at least $1 - Cn^{-p}$,

$$|Z_t - Z_s| \le C\sqrt{(t-s)\log n} \qquad \text{for } 0 \le s < t \le n\,. \tag{4.2}$$

Lemma 4.2. *On the event in (4.2), with the constant C of (4.2), the following holds for every unit vector u and all integers $0 \le s < t \le n$. If $u \cdot \xi(X_j) \ge 0$ at every first departure time j with $s \le j < t$, then*

$$u \cdot (X_t - X_s) \le C\sqrt{(t-s)\log n}\,.$$

Proof. By (4.1), $u \cdot (X_t - X_s) = u \cdot (Z_t - Z_s) - \kappa \sum_{s \le j < t} I_j u \cdot \xi(X_j) \le |Z_t - Z_s|$, and (4.2) bounds the right side. □

The martingales of the functions $(q \cdot x - a)_+$ bound the number of sites $x \in A_n$ at which $q \cdot x$ exceeds a given level (Step 1 of the proof of Lemma 4.3); we bound them simultaneously for all vectors q in a finite deterministic set, because q will depend on the path. For $q \in \mathbb{R}^d$ and $a \in \mathbb{R}$, let $\mathcal{M}^{q,a}$ be the martingale $\mathcal{M}^f$ of Section 3.1 with $f(x) = (q \cdot x - a)_+$. If $|q| \le \Lambda_\Psi$, then one step changes $q \cdot X_j$ by at most Λ_Ψ, so the increments of $\mathcal{M}^{q,a}$ are at most C, and the bracket of $\mathcal{M}^{q,a}$ through time n is at most $C \sum_{x: q\cdot x > a - \Lambda_\Psi} \ell_n(x)$, because f is constant on x and its neighbors unless $q \cdot x > a - \Lambda_\Psi$. Fix $p > 0$. Lemma 3.2 and a union bound over the $O(n^{d+1})$ pairs (q, a) with $q = \Lambda_\Psi u_x$ for some $x \in \mathbb{Z}^d$ with $0 < |x| \le n$ and $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$ give, with probability at least $1 - Cn^{-p}$, simultaneously for all these pairs,

$$|\mathcal{M}_n^{q,a}| \le C\left(\sqrt{\log n \sum_{x: q\cdot x > a - \Lambda_\Psi} \ell_n(x)} + \log n\right). \tag{4.3}$$

Lemma 4.3. *Let $\alpha > 0$. There exists $C(d, \kappa, \Psi, \alpha, p) < \infty$ with the following property. Let $q \in \mathbb{R}^d$ satisfy $|q| \le \Lambda_\Psi$, and consider a path of the walk that satisfies (4.2), and (4.3) for this q and every $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$. Let x_* be one of the sites $X_0, \ldots, X_n$ with $q \cdot X_j \le T := q \cdot x_*$ for $0 \le j \le n$, and let $h > 0$ be such that $q \cdot \xi(X_j) \ge \alpha$ for every $0 \le j \le n$ with $q \cdot X_j > T - h$. For $b \ge 0$ let $L_b := 1 + \max\{\ell_n(x) : x \in \mathbb{Z}^d,\ q \cdot x \ge b\}$. Then every $b \ge 0$ satisfies*

$$T \le \max\{T - h, b\} + CL_b \log n\,. \tag{4.4}$$

Proof. Fix $b \ge 0$ and write $L = L_b$. For $a \in \mathbb{R}$, let $\mathcal{N}(a)$ be the number of sites $x \in A_n$ with $q \cdot x > a$, and let a_0 be the least element of $\Lambda_\Psi \mathbb{Z}$ not smaller than $\max\{T - h, b + \Lambda_\Psi\}$. Every site $x \in A_n$ with $q \cdot x > a_0 - \Lambda_\Psi$ has local time less than L, because $q \cdot x > b$, and every site $x \in A_n$ with $q \cdot x > a_0$ satisfies $q \cdot \xi(x) \ge \alpha$, because $q \cdot x > T - h$. We show that $T \le a_0 + CL \log n$; since $a_0 \le \max\{T - h, b\} + 2\Lambda_\Psi$ and $L \ge 1$, this proves (4.4).

Step 1: Comparison of neighboring levels. Let $a \in \Lambda_\Psi \mathbb{Z}$ with $a \ge a_0$, and apply Dynkin's formula to $\phi(x) := (q \cdot x - a)_+$. If $q \cdot x \ge a + \Lambda_\Psi$, then ϕ is linear on x and its neighbors, so the increment of $\phi(X)$ at a departure from x has conditional mean zero, except at the first departure, where its conditional mean is at most $-\kappa\alpha$. If $q \cdot x \le a - \Lambda_\Psi$, then ϕ vanishes on x and its neighbors. The remaining departures contribute at most C each and occur at sites with local time less than L. For $a > T$ the left side of (4.5) below vanishes, and for $a \le T$ we have $a \le \Lambda_\Psi n$, because $|x_*| \le n$, so (4.3) bounds the martingale term. Since $\phi(X_0) = 0$ and $\phi(X_n) \ge 0$, it follows that

$$\mathcal{N}(a + \Lambda_\Psi) \le CL[\mathcal{N}(a - \Lambda_\Psi) - \mathcal{N}(a + \Lambda_\Psi)] + C\sqrt{L\mathcal{N}(a - \Lambda_\Psi)\log n} + C \log n\,. \tag{4.5}$$

Step 2: Iteration over the levels. For $i \geq 0$, let $N_i := \mathcal{N}(a_0 - \Lambda_\Psi + 2i\Lambda_\Psi)$. By (4.5) and Young's inequality, $N_{i+1} \leq (1 - c/L)N_i + C\log n$, with $0 < c < 1$. Since $N_0 \leq n$, taking $i = \lceil L\log n/c\rceil$ and $\bar{a} := a_0 - \Lambda_\Psi + 2i\Lambda_\Psi$ gives

$$a_0 \leq \bar{a} \leq a_0 + CL\log n \qquad \text{and} \qquad \mathcal{N}(\bar{a}) \leq CL\log n\,. \tag{4.6}$$

Step 3: The bound on T. If $T \leq \bar{a} + \Lambda_\Psi$, then $T \leq a_0 + CL\log n$ by (4.6). Otherwise $T > 0$, so $q \neq 0$. Let σ be the first time the walk visits x_*, and let $s := 1 + \max\{j < \sigma : q \cdot X_j \leq \bar{a}\}$; the maximum exists because $q \cdot X_0 = 0 \leq \bar{a}$. Each step changes $q \cdot X_j$ by at most Λ_Ψ, so $s < \sigma$ and $q \cdot X_s \leq \bar{a} + \Lambda_\Psi$. At the times $s \leq j < \sigma$ the walk departs from sites x with $q \cdot x > \bar{a}$. By (4.6) there are at most $CL\log n$ such sites, each with local time less than L, so $\sigma - s \leq CL^2\log n$; and $q \cdot \xi(X_j) \geq \alpha$ at every first departure in this interval. Lemma 4.2 with $u = q/|q|$ therefore gives

$$T - q \cdot X_s \leq C\sqrt{(\sigma - s)\log n} \leq CL\log n\,,$$

and hence $T \leq \bar{a} + \Lambda_\Psi + CL\log n \leq a_0 + CL\log n$ by (4.6). □

Let x_* be one of the sites $X_0, \ldots, X_n$ with $|x_*| = R_{\mathrm{out}}(n)$, so that $x_* \neq 0$ and $|x_*| \leq n$. Let $q := \Lambda_\Psi u_{x_*}$ and $T := q \cdot x_* = \Lambda_\Psi R_{\mathrm{out}}(n)$, so that $q \cdot X_j \leq \Lambda_\Psi |X_j| \leq T$ for $0 \leq j \leq n$, and let $\theta := c_\Psi^2/(8\Lambda_\Psi^2)$. If $q \cdot X_j > (1-\theta)T$, then $X_j \neq 0$ and $u_{x_*} \cdot u_{X_j} > (1-\theta)R_{\mathrm{out}}(n)/|X_j| \geq 1 - \theta$, so $|u_{x_*} - u_{X_j}| < (2\theta)^{1/2}$, and Euler's relation and $|\xi(X_j)| \leq \Lambda_\Psi$ give

$$q \cdot \xi(X_j) \geq \Lambda_\Psi\big(u_{X_j} \cdot \xi(X_j) - \Lambda_\Psi|u_{x_*} - u_{X_j}|\big) \geq \Lambda_\Psi\big(c_\Psi - \Lambda_\Psi(2\theta)^{1/2}\big) = \frac{\Lambda_\Psi c_\Psi}{2}\,.$$

Let $a \geq 0$. Every $x \in \mathbb{R}^d$ with $q \cdot x \geq \Lambda_\Psi a/c_\Psi$ satisfies $\Psi(x) \geq c_\Psi|x| \geq a$. So, on the event on which (4.2) and (4.3) hold, Lemma 4.3 with $h = \theta T$, $\alpha = \Lambda_\Psi c_\Psi/2$ and the level $\Lambda_\Psi a/c_\Psi$ in place of b gives

$$T \leq \max\Big\{(1-\theta)T, \frac{\Lambda_\Psi a}{c_\Psi}\Big\} + C\Big(1 + \max_{x \in \mathbb{Z}^d, \Psi(x) \geq a} \ell_n(x)\Big)\log n\,.$$

If the maximum is $(1-\theta)T$, then θT is at most the last term; if the maximum is $\Lambda_\Psi a/c_\Psi$, then $T - \Lambda_\Psi a/c_\Psi$ is at most the last term. Since $T = \Lambda_\Psi R_{\mathrm{out}}(n)$, in both cases, for every $a \geq 0$,

$$R_{\mathrm{out}}(n) \leq \frac{a}{c_\Psi} + C\Big(1 + \max_{x \in \mathbb{Z}^d, \Psi(x) \geq a} \ell_n(x)\Big)\log n\,. \tag{4.7}$$

For the Euclidean norm, $\xi(x) = u_x$ and $c_\Psi = 1$, and the maximum is over the sites with $|x| \geq a$.

4.2. The largest ball of the norm in the range.

The following objects are used in Sections 4–7. Let

$$b := \inf\{\Psi(y) : y \in \mathbb{R}^d \setminus D_n\} \qquad \text{and} \qquad E := D_n \setminus \{\Psi < b\}\,,$$

so that $\{\Psi < b\}$ is the largest ball of Ψ about the origin contained in D_n. Choose a point $y_0 \in \overline{\mathbb{R}^d \setminus D_n}$ with $\Psi(y_0) = b$, the **contact point**, and a cell C_z disjoint from D_n whose closure contains y_0. Then $\ell_n(z) = 0$ and $\Psi(z) \geq b$. Let

$$H := U_{D_n}(y_0) \qquad \text{and} \qquad \mathcal{I} := \int_E (\Psi(v) - b)\,\mathrm{d}v\,.$$

Lemma 2.2 gives $U_{\{\Psi<b\}}(y_0) = 0$, so $H = U_E(y_0)$. For almost every $v \in E$, the inequality (2.3) for the subgradient $\nabla\Psi(v)$ at v, with y_0 in place of y, gives

$$\nabla\Psi(v) \cdot (v - y_0) \geq \Psi(v) - b \geq 0\,. \tag{4.8}$$

Hence $H \geq 0$, and, if $E \subset B(y_0, \rho)$ for some $\rho > 0$, then

$$H \geq \frac{2\kappa}{\omega_d}\int_E \frac{\Psi(v) - b}{|v - y_0|^d}\,\mathrm{d}v \geq \frac{c\mathcal{I}}{\rho^d}\,, \qquad \text{so} \quad \mathcal{I} \leq C\rho^d H\,. \tag{4.9}$$

Since $\widetilde{\ell}_n$ vanishes on C_z and U_{D_n} is continuous,

$$0 \le H \le \sup_{y\in C_z} \left|\widetilde{\ell}_n(y) - U_{D_n}(y)\right| . \tag{4.10}$$

The integral $\mathcal{I}$ also bounds the potential of E. Let $a > 0$. The potential U_E is the sum of the potentials of the part of E where $\Psi \le b + a$ and of the rest of E. By Lemma 3.4, the potential of the first part is at most $2d\kappa a$ in absolute value; the rest of E has volume at most $\mathcal{I}/a$, because $\Psi - b > a$ there, so (3.14) bounds the absolute value of its potential by $C(\mathcal{I}/a)^{1/d}$. If $\mathcal{I} > 0$, then the choice $a \coloneqq \mathcal{I}^{1/(d+1)}$ gives

$$\sup_{y\in\mathbb{R}^d} |U_E(y)| \le C\mathcal{I}^{1/(d+1)} . \tag{4.11}$$

If $\mathcal{I} = 0$, then $|E| = 0$, because $E \subset \{\Psi \ge b\}$ and $\{\Psi = b\}$ has measure zero, so (4.11) holds as well.

4.3. Proof of the coarse bounds.

Proof of Proposition 4.1. For $R \ge 2$, let $\tau_R \coloneqq \inf\{t : \Psi(X_t) \ge R\}$. We first prove that, for every $p > 0$, with probability at least $1 - CR^{-p}$,

$$cR^{d+1} \le \tau_R \le CR^{d+1} \qquad \text{and} \qquad \max_x \ell_{\tau_R}(x) \le CR .$$

Step 1: Upper bounds at the exit time. The sites departed before time τ_R lie in $\{\Psi < R\} \subset B(0, R/c_\Psi)$. The sum of $(1 + |x - y|)^{1-d}$ over the sites x of a ball of radius CR is at most CR, uniformly in y, and the endpoint term in (3.6) is $O(\log R)$; so (3.6) and (3.5) give, for $t \le \tau_R$ and $y \in \mathbb{Z}^d$,

$$\ell_t(y) \le |\mathcal{M}^y_t| + CR .$$

Fix a constant C_0 with $C_0R/2 \ge R/c_\Psi + 2\sqrt{d}$ for every $R \ge 2$, so that $B(0, C_0R/2)$ contains every cell whose closure meets the closure of D_{τ_R}. Let γ_R be the first time t with $|\mathcal{M}^y_t| \ge R$ for some $y \in \mathbb{Z}^d \cap B(0, C_0R)$, and let $\sigma_R \coloneqq \tau_R \wedge \gamma_R$. For $t < \sigma_R$, every site of A_t has local time at most CR, so $t = \sum_x \ell_t(x) \le CR^{d+1}$. Thus $\sigma_R \le CR^{d+1}$ on every path, and $\max_x \ell_{\sigma_R}(x) \le CR$, because one step changes each local time by at most one. By (3.7),

$$\langle \mathcal{M}^y\rangle_{\sigma_R} \le CR \sum_{x\in B(0,R/c_\Psi)\cap\mathbb{Z}^d} (1 + |x - y|)^{2-2d} \le CR \log R .$$

Freedman's inequality [Freedman, 1975, Theorem 1.6] for the martingales $\mathcal{M}^y$ stopped at σ_R, with a union bound over the $O(R^d)$ sites y, shows that, except with probability at most CR^{-p},

$$\max_{y\in\mathbb{Z}^d\cap B(0,C_0R)} \max_{t\le\sigma_R} |\mathcal{M}^y_t| \le C\sqrt{R} \log R ,$$

which is less than R for large R. On this event $\gamma_R \le \tau_R$ is impossible, so $\sigma_R = \tau_R$, which proves the upper bounds on τ_R and on the local times at time τ_R.

Step 2: Lower bound on the exit time. We work on the event of Step 1. The replacements in Step 3 of the proof of Lemma 3.1 use only the locations of the departed sites, which lie in $B(0, R/c_\Psi)$; with the bound on the martingales in Step 1, they give

$$|\widetilde{\ell}_{\tau_R}(y) - U_{D_{\tau_R}}(y)| \le C\sqrt{R} \log R \qquad \text{for every } y \in \mathbb{R}^d \text{ with } |y| \le C_0R/2 .$$

Let b, E, y_0, C_z, H and $\mathcal{I}$ be the objects of Section 4.2 at time τ_R in place of n. The cell C_z lies in $B(0, C_0R/2)$, so (4.10) gives $H \le C\sqrt{R} \log R$. Since $D_{\tau_R} \subset B(0, CR)$, the bounds (4.9) and (4.11) give $\mathcal{I} \le CR^{d+1/2} \log R$ and $\sup |U_E| \le CR^{(2d+1)/(2d+2)} \log R$. By Lemma 2.2, $U_{D_{\tau_R}} = 2d\kappa(b - \Psi)_+ + U_E$, and both $\widetilde{\ell}_{\tau_R}$ and $2d\kappa(b - \Psi)_+$ vanish outside $B(0, C_0R/2)$; hence

$$\sup_{y\in\mathbb{R}^d} \left|\widetilde{\ell}_{\tau_R}(y) - 2d\kappa(b - \Psi(y))_+\right| \le CR^{(2d+1)/(2d+2)} \log R . \tag{4.12}$$

In particular, every site x with $\Psi(x) \ge b$ satisfies $\ell_{\tau_R}(x) \le CR^{(2d+1)/(2d+2)} \log R$.

Apply (4.7) at each deterministic time t with $R/\Lambda_\Psi \le t \le CR^{d+1}$, with $p+1$ in place of p; the failure probabilities sum to at most $C\sum_{t\ge R/\Lambda_\Psi} t^{-p-1} \le CR^{-p}$. Since $\Psi(X_t) \le \Lambda_\Psi t$, the time τ_R lies in this range on the event of Step 1, so (4.7) holds at time τ_R. With $a=b$ it gives

$$\frac{R}{\Lambda_\Psi} \le R_{\text{out}}(\tau_R) \le \frac{b}{c_\Psi} + CR^{(2d+1)/(2d+2)}(\log R)^2\,,$$

so $b \ge cR$ for large R. The two functions in (4.12) vanish outside D_{τ_R}, whose volume is at most CR^d, and their integrals are τ_R and $2d\kappa|B_\Psi|b^{d+1}/(d+1)$. Integrating (4.12) therefore gives

$$\tau_R \ge \frac{2d\kappa|B_\Psi|}{d+1}b^{d+1} - CR^{d+(2d+1)/(2d+2)}\log R \ge cR^{d+1}$$

for large R. This proves the bounds on τ_R.

Step 3: The bounds at time n. Take $R = C_1 r_n$, with the constant C_1 so large that $cR^{d+1} > n$, and apply the bounds on τ_R with $(d+1)(p+1)$ in place of p. Except with probability at most Cn^{-p}, we have $n < \tau_R$, and hence $R_{\text{out}}(n) \le R/c_\Psi \le Cr_n$ and $\max_x \ell_n(x) \le Cr_n$. It follows that $|A_n| \le C(1+R_{\text{out}}(n))^d \le Cr_n^d$. The identity $n = \sum_x \ell_n(x) \le |A_n| \max_x \ell_n(x)$ then gives $|A_n| \ge cr_n^d$ and $\max_x \ell_n(x) \ge cr_n$, and $|A_n| \le C(1+R_{\text{out}}(n))^d$ gives $R_{\text{out}}(n) \ge cr_n$. Enlarging the constants covers the remaining bounded values of n. □

5. The limit shape

We prove Theorem 2.1 with rates (Proposition 5.1). Let b, y_0 and C_z be as in Section 4.2, so that $\{\Psi < b\}$ is the largest ball of Ψ about the origin contained in D_n and y_0 is the contact point. By convexity, every point v of D_n outside this ball contributes at least $c(\Psi(v)-b)r_n^{-d}$ to $U_{D_n}(y_0)$. With the coarse bounds, Dynkin's formula at the center z of the cell C_z, a site outside A_n, shows that $U_{D_n}(y_0)$ is at most $C\sqrt{r_n \log n}$ if $d=2$ and at most $C\log n$ if $d\ge 3$ (Lemma 5.2). So little of D_n lies outside the ball, and the martingale obtained from $|X_n|^2$ in (5.5) shows that b is close to r_n.

For the outer bound we use Lemma 4.3 again. Let $x \mapsto q\cdot x$ be a linear function, and suppose that the first departure from every visited site where $q\cdot x$ is close to its maximum over $X_0,\dots,X_n$ decreases $q\cdot x$ in conditional mean. Then few sites of the range lie where $q\cdot x$ is large, and the walk cannot cross them to reach the maximum. When Ψ has corners, this hypothesis can fail for $q=\xi(x_*)$ at a site x_* where Ψ is largest: at a visited site y where $q\cdot y$ is close to $q\cdot x_*$, the drift $-\kappa\xi(y)$ can be orthogonal to $\xi(x_*)$. So we take a visited site x_* farthest from the ball $\{\Psi \le r_n\}$ in Euclidean distance, and we take q in the direction from the point of this ball nearest to x_* toward x_*.

Proposition 5.1. *Let Ψ be a norm, let $\xi(x)\in\partial\Psi(x)$ be subgradients chosen at the sites $x\in\mathbb{Z}^d\setminus\{0\}$, and let $\kappa>0$ satisfy (2.2). For every $p>0$ there exists $C(d,\kappa,\Psi,p)<\infty$, not depending on the choice of the subgradients, such that, for every integer $n\ge 2$, with probability at least $1-Cn^{-p}$ the following hold simultaneously.*

(i) Radii*:*

$$\left|\inf_{y\notin D_n}\Psi(y) - r_n\right| \le C\begin{cases} r_n^{3/4}(\log n)^{1/4}, & d=2,\\ (r_n\log n)^{1/2}, & d\ge 3\end{cases} \tag{5.1}$$

and

$$\max_{0\le j\le n}\Psi(X_j) - r_n \le C\begin{cases} r_n^{5/6}(\log n)^{7/6}, & d=2,\\ r_n^{d/(d+1)}(\log n)^{(d+2)/(d+1)}, & d\ge 3\end{cases}. \tag{5.2}$$

(ii) Local times*:*

$$\max_{x\in\mathbb{Z}^d}|\ell_n(x) - 2d\kappa(r_n-\Psi(x))_+| \le C\begin{cases} r_n^{5/6}(\log n)^{1/6}, & d=2,\\ r_n^{d/(d+1)}(\log n)^{1/(d+1)}, & d\ge 3\end{cases}. \tag{5.3}$$

Throughout the rest of the paper let

$$q_n := \begin{cases} \sqrt{\dfrac{\log n}{r_n}}, & d = 2, \\ \dfrac{\log n}{r_n}, & d \geq 3 \end{cases} . \tag{5.4}$$

The bounds in (5.1), (5.2) and (5.3) are $Cr_nq_n^{1/2}$, $Cr_nq_n^{1/(d+1)}\log n$ and $Cr_nq_n^{1/(d+1)}$.

5.1. **Martingale estimates.** We collect on one event the martingale bounds used in Sections 5–7. Fix $p > 0$. Since every step has squared length one and conditional mean $-\kappa I_j\xi(X_j)$ given $\mathcal{F}_j$, the process

$$\mathcal{Q}_t := \sum_{j<t} 2X_j \cdot \big(X_{j+1} - X_j + \kappa I_j \xi(X_j)\big)$$

is a martingale, and Euler's relation gives

$$|X_n|^2 = n - 2\kappa \sum_{x \in A_n} \Psi(x) + \mathcal{Q}_n\,. \tag{5.5}$$

Azuma's inequality for $\mathcal{Q}$ stopped on leaving $B(0,k)$, followed by a union bound over $1 \leq k \leq n+1$, gives, with probability at least $1 - Cn^{-p}$,

$$|\mathcal{Q}_n| \leq C\big(R_{\text{out}}(n) + 1\big)\sqrt{n \log n}\,. \tag{5.6}$$

We also use the martingales $\mathcal{M}^{q,a}$ of Section 4.1 for vectors q in the directions of Euclidean projections. For $x \in \mathbb{R}^d$, let $\Pi_n(x)$ be the point of the closed convex set $\{\Psi \leq r_n\}$ nearest to x in Euclidean distance. Lemma 3.2 and a union bound over the $O(n^{d+1})$ pairs (q,a) with $q = \Lambda_\Psi u_{x-\Pi_n(x)}$ for some $x \in \mathbb{Z}^d$ with $|x| \leq n$ and $\Psi(x) > r_n$, and $a \in \Lambda_\Psi\mathbb{Z} \cap [0, \Lambda_\Psi n]$, give (4.3) also for these pairs, with probability at least $1 - Cn^{-p}$.

Finally, for $y \in \mathbb{Z}^d$ let N^y be the martingale $\mathcal{M}^f$ of Section 3.1 for the squared Green function $f = (g_y - \gamma)^2$, where $\gamma := 0$ if $d \geq 3$ and $\gamma := (2/\pi)\log r_n$ if $d = 2$; these martingales are used in the proof of Lemma 5.2. By (3.4) and (3.5), if $|y| \leq 3n$, then the increments of N^y through time n are at most C if $d \geq 3$ and at most $C\log n$ if $d = 2$. Lemma 3.2 for N^y divided by this bound, with a union bound over the $O(n^d)$ sites y with $|y| \leq 3n$, gives an event of probability at least $1 - Cn^{-p}$.

Intersect this event and the events of Proposition 4.1 and Lemma 3.1 with the events on which (4.2), (3.13) and (5.6) hold and on which (4.3) holds for the pairs of Section 4.1 and for those above. This intersection is the event of Section 5.1; its complement has probability at most Cn^{-p}, and the arguments of Sections 5–7 are deterministic on it. Let K be a constant with $R_{\text{out}}(n) \leq (K-1)r_n$ on the event of Proposition 4.1; then $D_n \subset B(0, Kr_n)$ for large n, and (5.6) gives

$$|\mathcal{Q}_n| \leq Cr_n\sqrt{n \log n}\,. \tag{5.7}$$

By Proposition 4.1, $\max_x \ell_n(x) \leq Cr_n$, so (3.3) gives, for $y \in \mathbb{R}^d$ with $|y| \leq 2n$,

$$|\widetilde{\ell}_n(y) - U_{D_n}(y)| \leq Cr_n^{1/2}\log n\,. \tag{5.8}$$

5.2. **The inner radius and the local times.** Let b, E, y_0, z, H and $\mathcal{I}$ be as in Section 4.2. Then $|v - y_0| \leq Cr_n$ for $v \in E$, because $D_n \subset B(0, Kr_n)$ and $\Psi(y_0) = b \leq Cr_n$, so (4.9) gives $\mathcal{I} \leq Cr_n^dH$. By (4.10) and (5.8), $H \leq Cr_n^{1/2}\log n$.

Lemma 5.2 improves this bound, because the walk has not departed from z: up to $C\log n$, Dynkin's formula at z bounds H by $|\mathcal{M}_n^z|$. The bracket of $\mathcal{M}^z$ is at most a sum of the local times weighted by the squared increments of g_z. Dynkin's formula for the square of g_z, shifted by the constant γ in the plane, expresses this sum through $\ell_n(z) = 0$, endpoint terms, one term for each first departure, and the martingale N^z, whose bracket is controlled by the same sum. The same argument at every site y bounds $|\ell_n(y) - U_{D_n}(y)|$ in terms of $\ell_n(y)$; we use this bound again in Sections 6 and 7.

Lemma 5.2. *There exists $C(d, \kappa, \Psi, p) < \infty$ such that, on the event of Section 5.1, for large n,*

$$H \le C r_n q_n = C \begin{cases} \sqrt{r_n \log n}, & d = 2, \\ \log n, & d \ge 3 \end{cases} .$$

Proof. Step 1: Dynkin's formula for a squared Green function. For $y \in \mathbb{Z}^d$ let

$$\mathcal{E}_y := \frac{1}{2d} \sum_{x \in \mathbb{Z}^d} \ell_n(x) \sum_{|e|=1} \big(g_y(x+e) - g_y(x)\big)^2 .$$

Each step is to a neighbor, so at a departure from x the conditional variance of the increment of $\mathcal{M}^y$ is at most $\sum_{|e|=1} (g_y(x+e) - g_y(x))^2$; hence $\langle \mathcal{M}^y \rangle_n \le 2d\mathcal{E}_y$. Let $f_y := (g_y - \gamma)^2$, with γ as in Section 5.1, so that $N^y = \mathcal{M}^{f_y}$. Since $\Delta g_y = \mathbf{1}_{\{y\}}$, expanding the squares gives

$$\Delta f_y(x) = \frac{1}{2d} \sum_{|e|=1} \big(g_y(x+e) - g_y(x)\big)^2 + 2\big(g(0) - \gamma\big) \mathbf{1}_{\{y\}}(x) ,$$

and Dynkin's formula for f_y, with the first departures grouped by site as in (3.6), gives

$$\mathcal{E}_y = 2\big(\gamma - g(0)\big)\ell_n(y) + f_y(X_n) - f_y(0) - N^y_n + \kappa \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} f_y(x) .$$

The last sum has one term for each of the at most $C r_n^d$ sites of A_n. We take $y \in \mathbb{Z}^d$ with $|y| \le 3n$; on the event of Section 5.1, the bound of Lemma 3.2 for N^y and the bound (3.13) for $\mathcal{M}^y$ hold simultaneously for all these y.

Step 2: Higher dimensions. Let $d \ge 3$, so that $\gamma = 0$. Since $|g(v)| \le C(1 + |v|)^{2-d}$ and $|g(v+e) - g(v)| \le C(1+|v|)^{1-d}$ by (3.4) and (3.5), we have $|g(v+e)^2 - g(v)^2| \le C(1+|v|)^{3-2d}$ for every unit vector e. The sum of $(1 + |x-y|)^{3-2d}$ over at most $C r_n^d$ sites x is largest for the sites nearest to y, so it is $O(\log r_n)$ when $d = 3$ and $O(1)$ when $d \ge 4$; hence the last sum in the formula for $\mathcal{E}_y$ is at most $C \log n$. The terms $f_y(X_n)$ and $f_y(0)$ and the increments of N^y are bounded, and $\langle N^y \rangle_n \le C\mathcal{E}_y$, because $(g(v+e)^2 - g(v)^2)^2 \le C(g(v+e) - g(v))^2$. So Lemma 3.2 for N^y gives $\mathcal{E}_y \le C\ell_n(y) + C \log n + C(\mathcal{E}_y \log n)^{1/2}$, and hence $\mathcal{E}_y \le C(\ell_n(y) + \log n)$.

Step 3: The plane. Let $d = 2$, so that $\gamma = ({}^2\!/\!{}_\pi) \log r_n$. By (3.4) and (3.5), for every unit vector e the two factors of $(g(v+e) - \gamma)^2 - (g(v) - \gamma)^2 = (g(v+e) - g(v))(g(v+e) + g(v) - 2\gamma)$ are at most $C(1+|v|)^{-1}$ and $C(1 + |\log((1+|v|)/r_n)|)$ in absolute value. The product of these two bounds decreases in $|v|$, so its sum over at most $C r_n^2$ values $v = x - y$ is at most its sum over $|v| \le C r_n$, which is at most $C \int_1^{C r_n} (1 + |\log(s/r_n)|) \, ds \le C r_n$. Hence the last sum in the formula for $\mathcal{E}_y$ is at most $C r_n$. The terms $2(\gamma - g(0))\ell_n(y)$, $f_y(X_n)$ and $f_y(0)$ are at most $C\ell_n(y) \log n$, $C(\log n)^2$ and $C(\log n)^2$, the increments of N^y are at most $C \log n$, and $\langle N^y \rangle_n \le C(\log n)^2 \mathcal{E}_y$. So Lemma 3.2, applied to $N^y/(C \log n)$, and Young's inequality give

$$\mathcal{E}_y \le C\ell_n(y) \log n + C r_n + C(\log n)^2 + C(\log n)^{3/2} \mathcal{E}_y^{1/2} \le \frac{1}{2}\mathcal{E}_y + C\big(\ell_n(y) \log n + r_n + (\log n)^3\big) .$$

Since $(\log n)^3 = o(r_n)$, this gives $\mathcal{E}_y \le C(\ell_n(y) \log n + r_n)$.

Step 4: Local times and the potential. The bound (3.13) and $\langle \mathcal{M}^y \rangle_n \le 2d\mathcal{E}_y$ give $|\mathcal{M}^y_n| \le C((\mathcal{E}_y \log n)^{1/2} + \log n)$. With (3.11) and Steps 2 and 3, this gives, simultaneously for all $y \in \mathbb{Z}^d$ with $|y| \le 3n$,

$$|\ell_n(y) - U_{D_n}(y)| \le C \begin{cases} \sqrt{r_n \log n} + (\log n)\sqrt{\ell_n(y)}, & d = 2, \\ \log n + \sqrt{\ell_n(y) \log n}, & d \ge 3 \end{cases} . \tag{5.9}$$

We deduce that

$$U_{D_n} \geq \frac{1}{2}\widetilde{\ell}_n - Cr_nq_n \quad \text{on } \mathbb{R}^d \qquad \text{and} \qquad |U_{D_n}| \leq Cr_nq_n \quad \text{on } \mathbb{R}^d \setminus D_n\,. \tag{5.10}$$

At a site y with $|y| \leq 3n$, Young's inequality absorbs half of $\ell_n(y)$ on the right side of (5.9), because $(\log n)^2 = o(\sqrt{r_n \log n})$ when $d = 2$; this gives $U_{D_n}(y) \geq \ell_n(y)/2 - Cr_nq_n$, and $|U_{D_n}(y)| \leq Cr_nq_n$ if $y \notin A_n$. The function $\widetilde{\ell}_n$ is constant on cells, and the cells disjoint from D_n are those of the sites outside A_n, so (3.12) extends these bounds to $B(0, 2n)$. Outside $B(0, 2n)$ the local time vanishes and $|U_{D_n}| \leq C$, because $|D_n| \leq n$ and $D_n \subset B(0, n + \sqrt{d})$. This proves (5.10). Finally, y_0 lies in the closure of the cell C_z, which is disjoint from D_n, so the second bound in (5.10) and the continuity of U_{D_n} give $H = U_{D_n}(y_0) \leq Cr_nq_n$. □

Lemma 5.2 and the bound $\mathcal{I} \leq Cr_n^d H$ bound $\mathcal{I}$, hence $|E|$, and then (5.5) shows that b is close to r_n.

Proof of (5.1) *and* (5.3). We work on the event of Section 5.1 and take n large.

Step 1: The volume bound. Let $a := r_nq_n^{1/2}$. Split E into the shell $\{b \leq \Psi < b + a\}$, of volume at most $Cr_n^{d-1}a$, and its complement, of volume at most $\mathcal{I}/a$. Since $\mathcal{I} \leq Cr_n^d H$, Lemma 5.2 gives

$$|E| \leq Cr_n^{d-1}a + \frac{\mathcal{I}}{a} \leq Cr_n^{d-1}a + \frac{Cr_n^{d+1}q_n}{a} = Cr_n^d q_n^{1/2}\,. \tag{5.11}$$

Step 2: The inner radius. By (5.5), (5.7) and $|X_n| \leq R_{\mathrm{out}}(n) \leq Cr_n$,

$$\Big| n - 2\kappa \sum_{x \in A_n} \Psi(x) \Big| \leq Cr_n^{(d+3)/2}(\log n)^{1/2}\,.$$

Replacing the sum by $\int_{D_n} \Psi(v)\,\mathrm{d}v$ changes the sum by at most $\Lambda_\Psi(\sqrt{d}/2)|A_n| \leq Cr_n^d$. Also

$$\int_{D_n} \Psi(v)\,\mathrm{d}v = \frac{d|B_\Psi|}{d+1}b^{d+1} + \int_E \Psi(v)\,\mathrm{d}v \quad \text{and} \quad 0 \leq \int_E \Psi(v)\,\mathrm{d}v \leq Cr_n|E|\,.$$

Since $n = 2\kappa d|B_\Psi|r_n^{d+1}/(d+1)$ by (2.4), dividing by n gives

$$\left| 1 - \left(\frac{b}{r_n}\right)^{d+1} \right| \leq C\big(r_n^{(1-d)/2}(\log n)^{1/2} + r_n^{-1} + |E|r_n^{-d}\big)\,. \tag{5.12}$$

By (5.11), and because $r_n^{(1-d)/2}(\log n)^{1/2}$ and r_n^{-1} are at most $q_n^{1/2}$, the right side is at most $Cq_n^{1/2}$, so

$$|b - r_n| \leq Cr_nq_n^{1/2}\,. \tag{5.13}$$

Discarding $-|X_n|^2$ in (5.5) and $\int_E \Psi(v)\,\mathrm{d}v \geq 0$ before taking absolute values gives, without (5.11), the one-sided bound $(b - r_n)_+ \leq C(r_n^{(3-d)/2}(\log n)^{1/2} + 1)$.

Step 3: The local times. Lemma 2.2 gives $U_{D_n}(y) = 2d\kappa(b - \Psi(y))_+ + U_E(y)$ for every $y \in \mathbb{R}^d$, and (4.11), the bound $\mathcal{I} \leq Cr_n^d H$ and Lemma 5.2 give $\sup |U_E| \leq C(r_n^{d+1}q_n)^{1/(d+1)} = Cr_nq_n^{1/(d+1)}$. With (5.8) and (5.13), these give, for $|y| \leq 2n$,

$$|\widetilde{\ell}_n(y) - 2d\kappa(r_n - \Psi(y))_+| \leq Cr_n^{1/2}\log n + Cr_nq_n^{1/(d+1)} + 2d\kappa|b - r_n| \leq Cr_nq_n^{1/(d+1)}\,, \tag{5.14}$$

because $r_n^{1/2}\log n$ and $r_nq_n^{1/2}$ are at most $r_nq_n^{1/(d+1)}$; for $|y| > 2n$ both $\widetilde{\ell}_n(y)$ and $2d\kappa(r_n - \Psi(y))_+$ vanish. □

5.3. **The outer radius.** We bound $\max_{j\le n}\Psi(X_j)$ by Lemma 4.3 for a vector q parallel to $x_*-\Pi_n(x_*)$, where x_* is a visited site farthest from $\{\Psi\le r_n\}$ and Π_n is the Euclidean projection of Section 5.1. Recall that, for $x\in\mathbb{R}^d$, every y with $\Psi(y)\le r_n$ satisfies $(x-\Pi_n(x))\cdot(y-\Pi_n(x))\le 0$.

Proof of Proposition 5.1. We work on the event of Section 5.1 and take n large. Then (5.13), in which $b=\inf_{y\notin D_n}\Psi(y)$, gives (5.1), and, since $\widetilde{\ell}_n=\ell_n$ at lattice sites, (5.14) gives (5.3).

For the outer radius, let $w\coloneqq\max_{0\le j\le n}|X_j-\Pi_n(X_j)|$. If $w=0$, then $\Psi(X_j)\le r_n$ for $0\le j\le n$, and (5.2) holds. Otherwise, let x_* be one of the sites $X_0,\dots,X_n$ with $|x_*-\Pi_n(x_*)|=w$, and let

$$u\coloneqq\frac{x_*-\Pi_n(x_*)}{w}\,,\qquad q\coloneqq\Lambda_\Psi u\qquad\text{and}\qquad\beta\coloneqq q\cdot\Pi_n(x_*)\,.$$

The projection inequality gives $q\cdot y\le\beta$ on $\{\Psi\le r_n\}$, with strict inequality in its interior. In particular, $\beta\ge 0$, and

$$\Psi(x)\ge r_n\qquad\text{for every } x\in\mathbb{R}^d \text{ with } q\cdot x\ge\beta\,. \tag{5.15}$$

Let $T\coloneqq q\cdot x_*=\beta+\Lambda_\Psi w$. Since $|X_j-\Pi_n(X_j)|\le w$ and $|q|=\Lambda_\Psi$, we have $q\cdot X_j\le\beta+\Lambda_\Psi w=T$ for $0\le j\le n$.

Let $\theta\coloneqq c_\Psi^2/(8\Lambda_\Psi^2)$. We show that the drift hypothesis of Lemma 4.3 holds with $h=\theta\Lambda_\Psi w$ and $\alpha=\Lambda_\Psi c_\Psi/2$. Let y be one of the sites $X_0,\dots,X_n$ with $q\cdot y>T-\theta\Lambda_\Psi w$, and let $\xi\in\partial\Psi(y)$. Then $q\cdot y>\beta$, so $\Psi(y)>r_n$, because $\Psi(y)\le r_n$ would give $q\cdot y\le\beta$. Hence $w_y\coloneqq|y-\Pi_n(y)|$ is positive; let $v_y\coloneqq(y-\Pi_n(y))/w_y$. Since $q\cdot\Pi_n(y)\le\beta$, we get $\Lambda_\Psi w_y u\cdot v_y\ge q\cdot y-\beta>(1-\theta)\Lambda_\Psi w$, and since $w_y\le w$, this gives $u\cdot v_y>1-\theta$, so $|u-v_y|<(2\theta)^{1/2}$. The subgradient inequality at y and $\Psi(\Pi_n(y))\le r_n$ give $\xi\cdot(y-\Pi_n(y))\ge\Psi(y)-r_n$. Since $\Psi(r_n y/\Psi(y))=r_n$ and w_y is the distance from y to $\{\Psi\le r_n\}$, also $w_y\le|y|(1-r_n/\Psi(y))\le(\Psi(y)-r_n)/c_\Psi$. Hence $\xi\cdot v_y\ge c_\Psi$, and, because $|\xi|\le\Lambda_\Psi$,

$$q\cdot\xi\ge\Lambda_\Psi\bigl(\xi\cdot v_y-\Lambda_\Psi|u-v_y|\bigr)\ge\Lambda_\Psi\bigl(c_\Psi-\Lambda_\Psi(2\theta)^{1/2}\bigr)=\frac{\Lambda_\Psi c_\Psi}{2}\,.$$

By (5.15) and (5.14), every $x\in\mathbb{Z}^d$ with $q\cdot x\ge\beta$ satisfies $\ell_n(x)\le Cr_nq_n^{1/(d+1)}$. Since $x_*\in\mathbb{Z}^d$, $|x_*|\le n$ and $\Psi(x_*)>r_n$, the bound (4.3) holds for q and every $a\in\Lambda_\Psi\mathbb{Z}\cap[0,\Lambda_\Psi n]$; so Lemma 4.3 with $h=\theta\Lambda_\Psi w$, $\alpha=\Lambda_\Psi c_\Psi/2$ and β in place of b gives

$$T\le\max\{T-\theta\Lambda_\Psi w,\beta\}+Cr_nq_n^{1/(d+1)}\log n\,.$$

Since $T-\theta\Lambda_\Psi w=\beta+(1-\theta)\Lambda_\Psi w>\beta$, this gives $\theta\Lambda_\Psi w\le Cr_nq_n^{1/(d+1)}\log n$. Finally, $\Psi(X_j)\le\Psi(\Pi_n(X_j))+\Lambda_\Psi|X_j-\Pi_n(X_j)|\le r_n+\Lambda_\Psi w$ for $0\le j\le n$, which proves (5.2). ☐

Proof of Theorem 2.1. Apply Proposition 5.1 with $p>1$ and the Borel–Cantelli lemma. Its errors are $o(r_n)$, so the inner- and outer-radius bounds give (i), and the local-time bound gives (ii). At every fixed site x, part (ii) gives $\ell_n(x)/r_n\to 2d\kappa>0$, proving (iii). Taking the Euclidean norm gives Theorem 1.1. ☐

The proofs in Sections 2–5 use the symmetry $\Psi(-x)=\Psi(x)$ only to bound the coordinates of the subgradients. Let $B\subset\mathbb{R}^d$ be a compact convex set with the origin in its interior, and let $\Psi_B(x)\coloneqq\inf\{t>0: x\in tB\}$ be its Minkowski functional, which is convex and positively homogeneous and satisfies $c|x|\le\Psi_B(x)\le C|x|$. Every $\xi\in\partial\Psi_B(x)$ satisfies $-\Psi_B(-e_i)\le\xi_i\le\Psi_B(e_i)$; so, if $\kappa\max\{\Psi_B(e_i),\Psi_B(-e_i)\}<1/d$ for $1\le i\le d$, then the probabilities (2.1) are positive, and the proofs of Theorem 2.1 and Proposition 5.1 apply without change to Ψ_B, with $|B|$ in place of $|B_\Psi|$. Thus every such set B is the limit shape of the centrally excited random walk with Ψ_B in place of Ψ.

6. THE INNER RADIUS FOR THE EUCLIDEAN NORM

We prove the bounds on $|R_{\text{in}}(n) - r_n|$ and on the volume in Theorem 1.2 (i) and (ii), and a first uniform bound on the local times (Proposition 6.1). For the Euclidean norm the contact point y_0 lies on the sphere $\{|v| = R_{\text{in}}(n)\}$, and every point v of D_n outside the ball $B(0, R_{\text{in}}(n))$ contributes at least $c|v - y_0|^{2-d}/r_n$ to the potential at y_0 (inequality (6.3) below and Figure 5). Unlike the lower bound $c(\Psi(v) - b)r_n^{-d}$ of Section 5, this bound does not vanish near the sphere, so Lemma 5.2 bounds the volume of D_n outside the ball, and not only the integral $\mathcal{I}$.

Proposition 6.1 (Inner radius)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d, \kappa, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$, the following hold simultaneously, where q_n is defined in (5.4).*

(i) Inner radius*:*

$$|R_{\text{in}}(n) - r_n| \leq C r_n q_n \,. \tag{6.1}$$

(ii) Volume*:*

$$\left|r_n^{-1} D_n \triangle B(0,1)\right| \leq C q_n \,.$$

(iii) Local times*:*

$$\sup_{y \in \mathbb{R}^d} \left|\widetilde{\ell}_n(y) - 2d\kappa(r_n - |y|)_+\right| \leq C r_n q_n^{1/d} \,.$$

Fix $p > 0$. In this section and the next we take n large and work on the event of Section 5.1; its complement has probability at most Cn^{-p}. With K as in Section 5.1, we have $D_n \subset B(0, Kr_n)$, and on this ball (3.3) and $\max_x \ell_n(x) \leq C r_n$ give

$$|\widetilde{\ell}_n(y) - U_{D_n}(y)| \leq C \begin{cases} \sqrt{r_n} \log n, & d = 2, \\ \sqrt{r_n \log n}, & d \geq 3 \end{cases} . \tag{6.2}$$

Let $b := R_{\text{in}}(n)$ and $E := D_n \setminus B(0, b)$, and let y_0 and H be as in Section 4.2 for the Euclidean norm, so that $|y_0| = b$ and $H = U_{D_n}(y_0) = U_E(y_0)$.

6.1. The volume outside the inner ball.

If $s = |v| \geq b = |y_0|$, then

$$\frac{u_v \cdot (v - y_0)}{|v - y_0|^d} = \frac{s^2 - v \cdot y_0}{s|v - y_0|^d} \geq \frac{1}{2s}|v - y_0|^{2-d} \geq 2^{1-d} s^{1-d} \,. \tag{6.3}$$

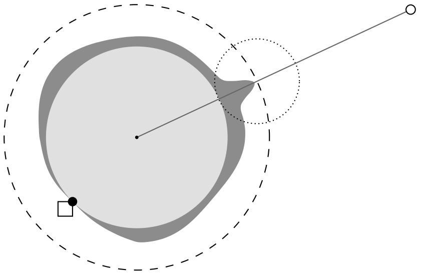

FIGURE 5. The light disk is the largest ball $B(0, b)$ about the origin in D_n, and the dark region is $E = D_n \setminus B(0, b)$. Every point of E contributes nonnegatively to $U_E(y_0)$ at the contact point y_0 (black dot), which lies on the circle $\{|v| = b\}$ and in the closure of a cell disjoint from D_n (square). The dashed circle, of radius $b + w = R_{\text{out}}(n) + 2\sqrt{d}$, contains D_n, and the points of E in the dotted ball $B((b + w)u, t)$ contribute negatively to $U_E(y_t)$ at the point $y_t = (b + w + 4t)u$ (open dot) on the segment from the origin (small dot) in the direction of a unit vector u.

The first inequality uses $2(s^2 - v \cdot y_0) = |v - y_0|^2 + s^2 - b^2$, and the second $|v - y_0| \le 2s$ and $d \ge 2$. Since $|v| \le Cr_n$ on E, integrating (6.3) over E gives $H \ge c|E| r_n^{1-d}$, and Lemma 5.2 gives

$$|E| \le C r_n^{d-1} H \le C r_n^d q_n \,. \tag{6.4}$$

6.2. **Upper bounds on the local times.** For a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, the potential of D with its negative contributions discarded is

$$U_D^+(y) := \frac{2\kappa}{\omega_d} \int_D \left[u_v \cdot \frac{v - y}{|v - y|^d} \right]_+ \mathrm{d}v \,.$$

The first bound in (5.10) gives $\widetilde{\ell}_n \le 2U_{D_n} + Cr_n q_n$. By (1.10), $U_{D_n} = U_{B(0,b)} + U_E$ and $U_{B(0,b)}$ vanishes outside $B(0,b)$; since $U_E \le \min\{U_E^+, \sup |U_E|\}$, every $y \in \mathbb{R}^d$ with $|y| \ge b$ satisfies

$$\widetilde{\ell}_n(y) \le 2 \min\bigl\{U_E^+(y), \sup |U_E|\bigr\} + C r_n q_n \,. \tag{6.5}$$

Let $L := 1 + \max\{\ell_n(x) : x \in \mathbb{Z}^d,\ |x| \ge b\}$. By (3.14) and (6.4), $\sup |U_E| \le C|E|^{1/d} \le C r_n q_n^{1/d}$, and $r_n q_n \le r_n q_n^{1/d}$; so (6.5) gives

$$L \le C r_n q_n^{1/d} \,. \tag{6.6}$$

6.3. The inner radius.

Proof of Proposition 6.1. By (5.12) and (6.4),

$$\left| 1 - \left(\frac{b}{r_n} \right)^{d+1} \right| \le C\bigl(r_n^{(1-d)/2} (\log n)^{1/2} + r_n^{-1} + q_n \bigr) \le C q_n \,,$$

because $r_n^{(1-d)/2}(\log n)^{1/2} \le q_n$. This proves (6.1). Since D_n is the disjoint union of $B(0,b)$ and E, the bounds (6.1) and (6.4) prove (ii).

On $B(0, Kr_n)$, the bounds (6.2), (1.10), (3.14) and (6.1) give

$$\bigl| \widetilde{\ell}_n(y) - 2d\kappa (r_n - |y|)_+ \bigr| \le C\bigl(\sqrt{r_n} \log n + r_n q_n + |E|^{1/d} \bigr) \le C r_n q_n^{1/d} \,,$$

and outside $B(0, Kr_n)$ both $\widetilde{\ell}_n(y)$ and $2d\kappa(r_n - |y|)_+$ vanish, which proves (iii). □

7. The outer radius and the local times

We prove the outer-radius and local-time bounds in Theorem 1.2. The crossing estimate (4.7) bounds the difference of the radii by $C \log n$ times one plus the largest local time outside the inner ball. To bound these local times, we bound the volume of the set $E = D_n \setminus B(0, R_{\mathrm{in}}(n))$ near the outer sphere, the sphere about the origin of radius $R_{\mathrm{out}}(n) + 2\sqrt{d}$. At a point beyond this sphere the potential is small, because the local time vanishes there. The points of E near that point contribute negatively to the potential, so a bound on the positive contributions bounds their volume (Figure 5). Because the outer sphere is curved, the positive contributions are bounded by the integral of $|v - y|^{2-d}/|v|$ over E. In higher dimensions we bound this integral by integrating the potential along rays; in the plane the volume bound (6.4) suffices.

Proposition 7.1 (Outer radius)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. For every $p > 0$ there exists $C(d, \kappa, p) < \infty$ such that, for every integer $n \ge 2$, with probability at least $1 - Cn^{-p}$,*

$$R_{\mathrm{out}}(n) \le r_n + C \begin{cases} \sqrt{r_n} (\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \ge 3 \end{cases} .$$

With $b = R_{\mathrm{in}}(n)$, $E = D_n \setminus B(0,b)$ and L as in Section 6, let

$$w := R_{\mathrm{out}}(n) - b + 2\sqrt{d}\,,$$

so that $w \geq 3\sqrt{d}/2 \geq 1$, because $b \leq R_{\mathrm{out}}(n) + \sqrt{d}/2$, and $E \subset \{b \leq |v| \leq b + w\}$. Then $b \asymp r_n$ and $b + w \asymp r_n$, by (6.1) and Proposition 4.1.

On the event of Section 5.1, the bound (4.7) with $a = b = R_{\mathrm{in}}(n)$ gives $R_{\mathrm{out}}(n) \leq b + CL \log n$. Because $L \geq 1$, this gives

$$w \leq CL \log n\,, \tag{7.1}$$

and with (6.6) the first bound

$$w \leq C r_n q_n^{1/d} \log n = o\big(r_n (\log n)^{-d}\big)\,. \tag{7.2}$$

7.1. **The potential beyond the inner radius.** Since E lies in the annulus $\{b \leq |v| \leq b + w\}$, Lemma 3.4 with $a = w$ gives

$$\sup_{y \in \mathbb{R}^d} |U_E(y)| \leq 2d\kappa w\,. \tag{7.3}$$

The following deterministic lemma bounds the positive contributions to the potential at a point between the two radii in terms of the volume near the sphere of radius $b + w$.

Lemma 7.2. *There exists $C(d,\kappa) < \infty$ with the following property. Let $1 \leq w \leq b$ and $\lambda \geq 1$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \leq |v| \leq b + w\}$ such that $|D \cap B((b+w)u, t)| \leq \lambda t^{d-1}$ for every unit vector u and every $t \geq w$, and $\int_D |v - y|^{2-d}\, \mathrm{d}v \leq \lambda b$ for every $y \in \mathbb{R}^d$. Then every $y \in \mathbb{R}^d$ with $b \leq |y| \leq b + w$ satisfies*

$$U_D^+(y) \leq C\big(\lambda w^{d-1}\big)^{1/d} + C\lambda\,.$$

Proof. Fix y with $b \leq |y| \leq b + w$. If $t \geq w$, then $B(y,t) \subset B((b+w)u_y, 2t)$, so $|D \cap B(y,t)| \leq 2^{d-1}\lambda t^{d-1}$. In particular, the points of D within distance w of y have total volume at most $2^{d-1}\lambda w^{d-1}$, so, as in the proof of (3.14), the integral of $|v - y|^{1-d}$ over them is at most $C(\lambda w^{d-1})^{1/d}$. For the other points $v \in D$, both $|v|$ and $|y|$ lie in $[b, b+w]$, so the identity $2(|v|^2 - v \cdot y) = |v - y|^2 + |v|^2 - |y|^2$ and $|v|^2 - |y|^2 \leq 3bw$ give

$$\left[u_v \cdot \frac{v - y}{|v - y|^d}\right]_+ \leq \frac{2w}{|v - y|^d} + \frac{1}{2b|v - y|^{d-2}}\,.$$

The second term contributes at most $C\lambda$ to $U_D^+(y)$, by the second hypothesis, and integrating the first by parts against the bound on $|D \cap B(y,t)|$ shows that it contributes at most $Cw \int_w^\infty \lambda t^{d-1} t^{-d-1}\, \mathrm{d}t \leq C\lambda$. □

We apply Lemma 7.2 to $D = E$. Its hypotheses come from the potential at points beyond the sphere of radius $b + w$. If $|y| \geq b + w$, then every $v \in E$ satisfies $|v| \leq |y|$, so the identity $2(|v|^2 - v \cdot y) = |v - y|^2 + |v|^2 - |y|^2$ gives $u_v \cdot (v - y) \leq |v - y|^2/(2|v|)$. Hence, with

$$F(y) := \int_E \frac{|v - y|^{2-d}}{|v|}\, \mathrm{d}v\,,$$

every $y \in \mathbb{R}^d$ with $|y| \geq b + w$ satisfies

$$U_E^+(y) \leq \frac{\kappa}{\omega_d} F(y)\,. \tag{7.4}$$

Sections 7.2 and 7.3 bound $\sup F$. Here we show that

$$w \leq C\lambda(\log n)^d\,, \qquad \text{where} \quad \lambda := r_n q_n + \sup_{\mathbb{R}^d} F\,. \tag{7.5}$$

Proof of (7.5). *Step 1: The volume of E near the sphere of radius $b+w$.* We show that, for every unit vector u and every $t \geq w$,

$$|E \cap B((b+w)u, t)| \leq Ct^{d-1}\lambda\,. \tag{7.6}$$

For $t > {r_n}/4$ this follows from (6.4), which gives $|E| \leq Cr_n^d q_n \leq Ct^{d-1} r_n q_n$. So let $w \leq t \leq {r_n}/4$, and let $y_t := (b+w+4t)u$ (Figure 5). Every point of D_n lies within distance $\sqrt{d}/2$ of a site x with $|x| \leq R_{\text{out}}(n) = b+w-2\sqrt{d}$, so $y_t \notin D_n$; also $U_{B(0,b)}(y_t) = 0$. The second bound in (5.10) therefore gives $|U_E(y_t)| = |U_{D_n}(y_t)| \leq Cr_n q_n$. If $v \in E \cap B((b+w)u, t)$, then $u_v \cdot u \geq 1 - t/(b+w)$, hence

$$u_v \cdot (v - y_t) = |v| - (b+w+4t)u_v \cdot u \leq (b+w) - (b+w+4t)\left(1 - \frac{t}{b+w}\right) \leq -t\,,$$

where the last inequality uses $t \leq {r_n}/4 \leq (b+w)/2$, which holds for large n by (6.1). Since also $|v - y_t| \leq 5t$, each such v satisfies $u_v \cdot (v-y_t)|v-y_t|^{-d} \leq -ct^{1-d}$. We bound $U_E(y_t)$ by the contributions of the points of $E \cap B((b+w)u,t)$ and the positive contributions of the other points of E; by (7.4),

$$-Cr_n q_n \leq U_E(y_t) \leq CF(y_t) - ct^{1-d}|E \cap B((b+w)u,t)|\,,$$

which proves (7.6).

Step 2: The bound on w. Since $|v| \leq b+w \leq 2b$ on E, every $y \in \mathbb{R}^d$ satisfies $\int_E |v-y|^{2-d}\,\mathrm{d}v \leq 2bF(y) \leq 2b\lambda$. With (7.6), this shows that Lemma 7.2 applies to $D = E$ with $C\lambda$ in place of λ; here $1 \leq w \leq b$ by (7.2), and $\lambda \geq r_n q_n \geq 1$. It gives, for every $y \in \mathbb{R}^d$ with $b \leq |y| \leq b+w$,

$$U_E^+(y) \leq C\big(\lambda w^{d-1}\big)^{1/d} + C\lambda\,. \tag{7.7}$$

Every site $x \in A_n$ with $|x| \geq b$ satisfies $|x| \leq b+w$, so (6.5) and (7.7) give $L \leq C((\lambda w^{d-1})^{1/d} + \lambda)$. With (7.1) and Young's inequality, this gives

$$w \leq C\log n\Big(\big(\lambda w^{d-1}\big)^{1/d} + \lambda\Big) \leq \frac{w}{2} + C\lambda(\log n)^d\,,$$

which proves (7.5). ☐

7.2. **The outer radius in higher dimensions.** Suppose that $d \geq 3$, so that $r_n q_n = \log n$. We prove that

$$\sup_{\mathbb{R}^d} F \leq C\log n\,. \tag{7.8}$$

Then $\lambda \leq C\log n$, and (7.5) gives

$$w \leq C(\log n)^{d+1}\,. \tag{7.9}$$

Proof of (7.8). Let u be a unit vector. The derivative of $t^{d-2}|v-tu|^{2-d}$ in $t>0$ is $(d-2)t^{d-3}v \cdot (v-tu)|v-tu|^{-d}$, so the definition of U_E gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(t^{d-2}F(tu)\big) = \frac{(d-2)\omega_d}{2\kappa}t^{d-3}U_E(tu)\,.$$

For $t \geq b$ we have $U_{B(0,b)}(tu) = 0$, so the first bound in (5.10) gives $U_E(tu) = U_{D_n}(tu) \geq -C\log n$. Also $|v - 2(b+w)u| \geq b+w$ and $|v| \geq b$ for $v \in E$, so (6.4) and $b \asymp r_n$ give $F(2(b+w)u) \leq |E|(b+w)^{2-d}/b \leq C\log n$. Integrating from t to $2(b+w)$ gives, for $b \leq t \leq 2(b+w)$,

$$t^{d-2}F(tu) \leq \big(2(b+w)\big)^{d-2}F\big(2(b+w)u\big) + C\log n\int_t^{2(b+w)} s^{d-3}\,\mathrm{d}s \leq Cr_n^{d-2}\log n\,,$$

so, since $t \geq b \geq cr_n$, we have $F \leq C\log n$ on $\{b \leq |y| \leq 2(b+w)\}$. The function F is continuous, harmonic off the closure of E, which lies in $\{b \leq |v| \leq b+w\}$, and tends to zero at infinity, so the maximum principle extends this bound to $B(0,b)$ and to the complement of $B(0,2(b+w))$. ☐

7.3. **The outer radius in the plane.** Suppose that $d = 2$, so that $r_n q_n = \sqrt{r_n \log n}$. Then $F(y) = \int_E |v|^{-1}\, dv \le |E|/b$ for every $y \in \mathbb{R}^2$, so (6.4) and $b \asymp r_n$ give $\sup F \le C\sqrt{r_n \log n}$. Hence $\lambda \le C\sqrt{r_n \log n}$, and (7.5) gives

$$w \le C\sqrt{r_n}(\log n)^{5/2}\,. \tag{7.10}$$

Proof of Proposition 7.1. On the event of Section 5.1, for large n, we have $R_{\text{out}}(n) \le b + w$ and, by (6.1), $b - r_n \le C r_n q_n$, which is $C \log n$ when $d \ge 3$ and $C\sqrt{r_n \log n}$ when $d = 2$. So (7.9) and (7.10) prove the proposition. ☐

7.4. **The uniform bound on the local times.** We prove Theorem 1.2 (iii) on the event of Section 5.1. Suppose first that $d \ge 3$. By (7.3) and (7.9), $\sup |U_E| \le Cw \le C(\log n)^{d+1}$, which with (6.2), (1.10) and (6.1) gives

$$\max_{y \in \mathbb{Z}^d, |y| \le K r_n} \bigl|\ell_n(y) - 2d\kappa(r_n - |y|)_+\bigr| \le C\bigl(\sqrt{r_n \log n} + (\log n)^{d+1} + |b - r_n|\bigr) \le C\sqrt{r_n \log n}$$

for large n, since $|b - r_n| \le C \log n$; outside $B(0, K r_n)$ both $\ell_n(y)$ and $2d\kappa(r_n - |y|)_+$ vanish.

Let $d = 2$. The function U_E is continuous (Section 3.2), harmonic in $B(0, b)$ because E does not meet that ball and each coordinate of $(v - y)|v - y|^{-2} = -\nabla_y \log |v - y|$ is harmonic in $y \ne v$, and nonnegative on the closure of $B(0, b)$ because $u_v \cdot (v - y) \ge 0$ when $|v| \ge b \ge |y|$. The maximum principle, the inequality $U_E \le U_E^+$, the bound (7.7) with $\lambda \le C\sqrt{r_n \log n}$ (Section 7.3) and (7.10) give

$$\sup_{|y| \le b + w} U_E(y) \le C\sqrt{r_n}(\log n)^{3/2}\,.$$

For $|y| \le b$, the bounds (6.2) and (1.10) give

$$\bigl|\widetilde{\ell}_n(y) - 2d\kappa(r_n - |y|)_+\bigr| \le C\bigl(\sqrt{r_n}(\log n)^{3/2} + |b - r_n|\bigr)\,.$$

For $|y| \ge b$, the absolute value of this difference is at most the larger of $\widetilde{\ell}_n(y)$ and $2d\kappa(r_n - |y|)_+$, which are nonnegative. Here $2d\kappa(r_n - |y|)_+ \le 2d\kappa|b - r_n|$, and $\widetilde{\ell}_n(y)$ is at most $U_E(y) + C\sqrt{r_n} \log n$ by (6.2) if $|y| \le b + w$ and vanishes otherwise. Since $|b - r_n| \le C\sqrt{r_n \log n}$, restricting to sites proves Theorem 1.2 (iii) for $d = 2$.

Proof of Theorem 1.2. On the event of Section 5.1, for large n, Proposition 6.1 gives the bounds on $R_{\text{in}}(n)$ and on the volume, Proposition 7.1 the bound on $R_{\text{out}}(n)$, and Section 7.4 the bound on the local times. The Borel–Cantelli lemma with $p > 1$ gives the almost-sure statement. ☐

8. LIMIT LAWS

We prove central limit theorems and laws of the iterated logarithm for $\sum_{x \in A_n} |x|$ and for the local times at fixed sites. After centering, these quantities are, up to smaller errors, linear combinations of the martingale $\mathcal{Q}$ of (5.5) and of the martingales $\mathcal{M}^y$ of Section 3.1; we compute the limits of their brackets from Theorem 1.1 (ii), and then apply the martingale central limit theorem and the law of the iterated logarithm.

8.1. **The sum of the distances from the origin.** For the Euclidean norm, (5.5) reads

$$\mathcal{Q}_n = 2\kappa \sum_{x \in A_n} |x| - n + |X_n|^2\,. \tag{8.1}$$

Let $R_{\text{mom}}(n)$ be the radius of the ball about the origin on which the integral of $|v|$ equals $\sum_{x \in A_n} |x|$, that is,

$$R_{\text{mom}}(n) := \left(\frac{d+1}{d\omega_d} \sum_{x \in A_n} |x|\right)^{1/(d+1)}\,. \tag{8.2}$$

Theorem 8.1 (Sum of the distances from the origin)**.** *Let Ψ be the Euclidean norm, let $0 < \kappa < 1/d$, and let $R_{\mathrm{mom}}(n)$ be as in (8.2). Then the following hold.*

(i) Central limit theorem*: as $n \to \infty$,*

$$\frac{\sum_{x \in A_n} |x| - n/(2\kappa)}{r_n^{(d+3)/2}} \xrightarrow{\mathrm{d}} \mathcal{N}\Big(0, \frac{2d\omega_d}{\kappa(d+2)(d+3)}\Big)$$

and

$$\frac{R_{\mathrm{mom}}(n) - r_n}{r_n^{(3-d)/2}} \xrightarrow{\mathrm{d}} \mathcal{N}\Big(0, \frac{2}{\kappa d\omega_d(d+2)(d+3)}\Big) .$$

(ii) Law of the iterated logarithm*: for each choice of sign, almost surely,*

$$\limsup_{n\to\infty} \frac{\pm\big(\sum_{x \in A_n} |x| - n/(2\kappa)\big)}{r_n^{(d+3)/2}\sqrt{2\log\log n}} = \Big(\frac{2d\omega_d}{\kappa(d+2)(d+3)}\Big)^{1/2},$$

$$\limsup_{n\to\infty} \frac{\pm(R_{\mathrm{mom}}(n) - r_n)}{r_n^{(3-d)/2}\sqrt{2\log\log n}} = \Big(\frac{2}{\kappa d\omega_d(d+2)(d+3)}\Big)^{1/2} . \tag{8.3}$$

In the plane, comparing $R_{\mathrm{mom}}(n)$ with the inner and outer radii gives Theorem 1.3 (ii) and, with a lower bound on the probability that $\mathcal{Q}_n$ deviates by about $(\log n)^{1/2}$ standard deviations (Lemma 9.1), Theorem 1.3 (i).

Proof of Theorem 8.1. Step 1: The bracket. Under both (2.1) and the simple random walk transition probabilities, the conditional second-moment matrix of $X_{j+1} - X_j$ is d^{-1} times the identity. Thus, for an $\mathcal{F}_j$-measurable vector q, the conditional variance of $q \cdot (X_{j+1} - X_j)$ is $|q|^2/d - \kappa^2 I_j (q \cdot u_{X_j})^2$. Since $\mathcal{Q}_{j+1} - \mathcal{Q}_j = 2X_j \cdot (X_{j+1} - X_j + \kappa I_j u_{X_j})$, this gives

$$\langle \mathcal{Q} \rangle_n = \frac{4}{d} \sum_{x \in \mathbb{Z}^d} \ell_n(x)|x|^2 - 4\kappa^2 \sum_{x \in A_n} |x|^2 . \tag{8.4}$$

Theorem 1.1 gives $R_{\mathrm{out}}(n) = O(r_n)$ and uniform convergence of the rescaled local times to $2d\kappa(1 - |v|)_+$ almost surely. Hence the first sum, divided by r_n^{d+3}, converges to

$$2d\kappa \int_{|v|<1} (1 - |v|)|v|^2 \,\mathrm{d}v = \frac{2d^2\kappa\omega_d}{(d+2)(d+3)} .$$

The second sum is $O(r_n^{d+2})$, so

$$\frac{\langle \mathcal{Q} \rangle_n}{r_n^{d+3}} \longrightarrow \frac{8d\kappa\omega_d}{(d+2)(d+3)} \qquad \text{almost surely} . \tag{8.5}$$

Step 2: The limit laws. The increment bound $|\mathcal{Q}_{j+1} - \mathcal{Q}_j| \le C|X_j|$ verifies the conditional Lindeberg condition at scale $r_n^{(d+3)/2}$. It also verifies Stout's condition, since

$$\frac{C|X_{n-1}|}{\sqrt{\langle \mathcal{Q} \rangle_n / \log\log \langle \mathcal{Q} \rangle_n}} = O\left(\sqrt{\frac{\log\log n}{n}}\right) \qquad \text{almost surely} .$$

Thus Hall and Heyde [1980, Corollary 3.1] and Stout [1970], applied to both signs, give the central limit theorem and the law of the iterated logarithm for $\mathcal{Q}$, with variance constant (8.5). In (8.1), the endpoint term $|X_n|^2 = O(r_n^2)$ is negligible on both scales, giving the assertions for the sum. These also give $R_{\mathrm{mom}}(n)/r_n \to 1$ almost surely. Expanding (8.2) now gives

$$\frac{R_{\mathrm{mom}}(n) - r_n}{r_n^{(3-d)/2}} = \Big(\frac{1}{d\omega_d} + o(1)\Big) \frac{\sum_{x \in A_n} |x| - n/(2\kappa)}{r_n^{(d+3)/2}} ,$$

which gives both assertions for $R_{\mathrm{mom}}(n)$. ☐

8.2. **The potential at a fixed site.** Since $B(0,R_{\mathrm{in}}(n))\subset D_n\subset B(0,R_{\mathrm{out}}(n)+\sqrt{d})$, Theorem 1.2 (i) and (ii) give, almost surely for all large n, $|D_n\triangle B(0,r_n)|\le Cr_n^d q_n$ and, for every $v\in D_n\triangle B(0,r_n)$,

$$\big||v|-r_n\big|\le C\begin{cases}\sqrt{r_n}(\log n)^{5/2}, & d=2,\\ (\log n)^{d+1}, & d\ge 3\end{cases}. \tag{8.6}$$

The product of q_n and the right side of (8.6) is $C(\log n)^3$ when $d=2$ and $C(\log n)^{d+2}/r_n$ when $d\ge 3$. On the annulus defined by (8.6), the kernel $|v|^{1-d}$ of $U_{D_n}(0)$ is close to $r_n^{-d}|v|$, which relates $U_{D_n}(0)$ to $\sum_{x\in A_n}|x|$.

Lemma 8.2. *Let Ψ be the Euclidean norm, let $0<\kappa<1/d$, and let $y\in\mathbb{Z}^d$. There exists $C(d,\kappa,y)<\infty$ such that, almost surely, for all sufficiently large n,*

$$\left|U_{D_n}(y)-2d\kappa(r_n-|y|)-\frac{\mathcal{Q}_n}{\omega_d r_n^d}\right|\le C\begin{cases}(\log n)^3, & d=2,\\ 1, & d\ge 3\end{cases}, \tag{8.7}$$

where $\mathcal{Q}$ is the martingale of (5.5).

By Theorem 8.1, $\mathcal{Q}_n/(\omega_d r_n^d)$ has order $r_n^{(3-d)/2}$, smaller than the order of $\mathcal{M}_n^y$ by a factor $(\log r_n)^{1/2}$ in the plane and $r_n^{(d-2)/2}$ when $d\ge 3$.

Proof. Write $r=r_n$, $\nu=\mathbf{1}_{D_n}-\mathbf{1}_{B(0,r)}$, and $w=\sup_{v\in\operatorname{supp}\nu}\big||v|-r\big|$. By (8.6), $w=o(r)$ and $\|\nu\|_1\le Cr^d q_n$. The function $s^{1-d}-r^{-d}s$ vanishes at r and has derivative $O(r^{-d})$ on $[r/2,\infty)$, so the definition of U_D gives

$$U_{D_n}(0)-2d\kappa r=\frac{2\kappa}{\omega_d r^d}\left(\int_{D_n}|v|\,\mathrm{d}v-\frac{n}{2\kappa}\right)+O(q_n w)\,.$$

Replacing the integral by $\sum_{x\in A_n}|x|$ costs $O(|A_n|)=O(r^d)$. By (8.1), the displayed main term is therefore $\mathcal{Q}_n/(\omega_d r^d)+O(1)$, since $|X_n|^2/r^d=O(1)$. The bounds following (8.6) give $q_n w=O((\log n)^3)$ in the plane and $O(1)$ in higher dimensions.

For fixed y, let $U_\nu=U_{D_n}-U_{B(0,r)}$. Differentiating its kernel with respect to y on $\operatorname{supp}\nu\subset\{|v|\ge r/2\}$ gives $|U_\nu(y)-U_\nu(0)|\le C_y r^{-d}\|\nu\|_1=O_y(1)$. Finally (1.10) gives $U_{B(0,r)}(y)-U_{B(0,r)}(0)=-2d\kappa|y|$. ☐

8.3. **Local times at fixed sites.** Up to a smaller correction at first departures, the bracket of $\mathcal{M}^y$ is a sum of local times with weights that decay like $|x-y|^{2-2d}$; in the plane each dyadic annulus about y inside $B(0,r_n)$ contributes order r_n, which gives an extra factor $\log r_n$. For functions $f_1,f_2\colon\mathbb{Z}^d\to\mathbb{R}$ with bounded nearest-neighbor increments, let

$$\Gamma(f_1,f_2)(x):=\frac{1}{2d}\sum_{|e|=1}\big(f_1(x+e)-f_1(x)\big)\big(f_2(x+e)-f_2(x)\big)-\Delta f_1(x)\Delta f_2(x)\,, \tag{8.8}$$

which is the conditional covariance of the increments of $\mathcal{M}^{f_1}$ and $\mathcal{M}^{f_2}$ at a departure from x other than the first. As in Section 3.1, $g_y=g(\cdot-y)$ and G is the Green function.

Theorem 8.3 (Local times at fixed sites)**.** *Let Ψ be the Euclidean norm, and let $0<\kappa<1/d$. Then the following hold.*

(i) Central limit theorem*: for every $y\in\mathbb{Z}^d$, as $n\to\infty$,*

$$\begin{aligned}\frac{\ell_n(y)-2d\kappa(r_n-|y|)}{\sqrt{r_n\log r_n}}&\xrightarrow{\mathrm{d}}\mathcal{N}\Big(0,\frac{16\kappa}{\pi}\Big) && \text{if } d=2\,,\\ \frac{\ell_n(y)-2d\kappa(r_n-|y|)}{\sqrt{r_n}}&\xrightarrow{\mathrm{d}}\mathcal{N}\big(0,2d\kappa(2G(0)-1)\big) && \text{if } d\ge 3\,.\end{aligned} \tag{8.9}$$

(ii) Law of the iterated logarithm*: for every $y \in \mathbb{Z}^d$ and each choice of sign, almost surely,*

$$\limsup_{n\to\infty} \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{\sqrt{2r_n \log r_n \log\log n}} = \Big(\frac{16\kappa}{\pi}\Big)^{1/2} \qquad \text{if } d = 2\,,$$
$$\limsup_{n\to\infty} \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{\sqrt{2r_n \log\log n}} = \big(2d\kappa(2G(0) - 1)\big)^{1/2} \qquad \text{if } d \geq 3\,.$$

(iii) Joint limits*: for all $y_1, \dots, y_k \in \mathbb{Z}^d$, the convergence in (8.9) holds jointly for $y = y_1, \dots, y_k$, and the limit is a centered Gaussian vector whose covariance matrix has (i,j) entry*

$$\begin{cases} 16\kappa/\pi, & d = 2, \\ 2d\kappa\big(2G(y_i - y_j) - \mathbf{1}_{\{y_i = y_j\}}\big), & d \geq 3 \end{cases}. \tag{8.10}$$

In the plane the limits at different sites are equal; this statement concerns finitely many fixed sites, whereas the maximum in Theorem 1.3 (iii) is over the sites x with $|x| \leq r_n^{1/2}$, a set that grows with n.

Proof of Theorem 8.3. Step 1: The bracket. A first departure from x changes the conditional covariance from $\Gamma(g_y, g_z)(x)$ by $O_{y,z}((1 + |x|)^{2-2d})$, by (3.5). There is one such departure per site of A_n, which is contained in $B(0, Cr_n)$ eventually almost surely. Thus

$$\langle \mathcal{M}^y, \mathcal{M}^z \rangle_n = \sum_{x \in \mathbb{Z}^d} \ell_n(x)\Gamma(g_y, g_z)(x) + \begin{cases} O_{y,z}(\log r_n), & d = 2, \\ O_{y,z}(1), & d \geq 3 \end{cases}. \tag{8.11}$$

Step 2: The bracket in higher dimensions. The identity $\Gamma(f, h) = \Delta(fh) - f\Delta h - h\Delta f - \Delta f \Delta h$ gives

$$\sum_x \Gamma(f, h)(x) = -\sum_x \big(f(x)\Delta h(x) + h(x)\Delta f(x) + \Delta f(x)\Delta h(x)\big)$$

whenever the boundary term vanishes. For $f = g_y$ and $h = g_z$ in $d \geq 3$ it is $O(R^{2-d})$ on a ball of radius R, by (3.4) and (3.5). Since $\Delta g_y = \mathbf{1}_{\{y\}}$, the sum is $2G(y - z) - \mathbf{1}_{\{y=z\}}$. The weights are absolutely summable, so Theorem 1.1 (ii) and dominated convergence give

$$\frac{\langle \mathcal{M}^y, \mathcal{M}^z \rangle_n}{r_n} \longrightarrow 2d\kappa\big(2G(y - z) - \mathbf{1}_{\{y=z\}}\big) \qquad \text{almost surely}\,.$$

Step 3: The bracket in the plane. The asymptotic formulas (3.4) and (3.5) give, for fixed y, z,

$$\Gamma(g_y, g_z)(x) = \frac{2}{\pi^2|x|^2} + O_{y,z}(|x|^{-3}), \qquad \sum_{|x| \leq r} \Gamma(g_y, g_z)(x) = \frac{4}{\pi}\log r + O_{y,z}(1)\,.$$

In (8.11), the uniform error $o(r_n)$ of Theorem 1.1 (ii) costs $o(r_n \log r_n)$. The term $4\kappa r_n$ contributes $(16\kappa/\pi) r_n \log r_n + O(r_n)$. The term $-4\kappa|x|$ contributes $O(r_n)$, since

$$\sum_{|x| \leq r_n} |x|\, |\Gamma(g_y, g_z)(x)| = O(r_n)\,.$$

Therefore

$$\frac{\langle \mathcal{M}^y, \mathcal{M}^z \rangle_n}{r_n \log r_n} \longrightarrow \frac{16\kappa}{\pi} \qquad \text{almost surely}\,.$$

Step 4: The limit laws. The bounded increments and these bracket limits verify the hypotheses of the martingale central limit theorem [Hall and Heyde, 1980, Corollary 3.1] and of Stout's law of the iterated logarithm [Stout, 1970]. The Cramér–Wold device gives the joint Gaussian limits for $(-\mathcal{M}_n^{y_i})_i$; a linear combination with zero limiting variance tends to zero in probability, by Chebyshev's inequality for the martingale stopped when its normalized bracket exceeds a small constant. Stout's law, applied to $\mathcal{M}^y$ and $-\mathcal{M}^y$, gives both signs of the almost-sure limits at each fixed site. Finally (3.11) and Lemma 8.2 give

$$\ell_n(y) - 2d\kappa(r_n - |y|) = -\mathcal{M}^y_n + \frac{\mathcal{Q}_n}{\omega_d r_n^d} + \begin{cases} O_y((\log n)^3), & d = 2, \\ O_y(\log n), & d \geq 3 \end{cases} .$$

Theorem 8.1 makes the last two terms negligible on both normalizations, proving all three assertions. □

9. Lower bounds on the fluctuations

We prove the lower bounds of Theorem 1.3, an upper bound on the deviations of the local times away from the boundary (Proposition 9.2), and logarithmic lower bounds on the radii (Proposition 9.4). The lower bounds in Theorem 1.3 (i), (iii) and (iv)(a) need martingale deviations of about $(\log n)^{1/2}$ standard deviations. We obtain them from exponential martingales (Lemma 9.1): for parts (i) and (iv)(a) by the Paley–Zygmund inequality for one exponential martingale, and for part (iii) by a second-moment bound for the average of the exponential martingales at many well-separated sites. Part (iii) needs many sites because, by Theorem 8.3, the deviation at one fixed site is typically smaller than the bound in part (iii) by a factor of order $(\log n)^{1/2}$.

Throughout this section $0 < \kappa < 1/d$, and Ω_n is the event of Section 5.1 with $p = 10$, so $\mathbb{P}(\Omega_n^c) = O(n^{-10})$ and the estimates of Sections 3–7 used below hold on Ω_n for large n. In particular, by Section 7.4, on Ω_n for large n, in every dimension,

$$\max_{x\in\mathbb{Z}^d} \big|\ell_n(x) - 2d\kappa(r_n - |x|)_+\big| \leq C\sqrt{r_n}(\log n)^{3/2} . \tag{9.1}$$

Lemma 9.1. *Let $0 < c_0 \leq C_0$. There exist $c(c_0, C_0) > 0$ and $C(c_0, C_0) < \infty$ with the following property. Let m and n be positive integers, let $b > 0$, $\delta \geq 0$ and $0 \leq \alpha \leq 1$, and let $S^1, \ldots, S^m$ be real martingales with respect to a common filtration, with $S^i_0 = 0$ and $|S^i_{t+1} - S^i_t| \leq b$ for $0 \leq t < n$. Let Ω be an event with $\mathbb{P}(\Omega) \geq 1 - \alpha$ on which $c_0 \leq \langle S^i\rangle_n \leq C_0$ for $1 \leq i \leq m$. Then, for every real $\beta \geq C$ with $\beta b \leq c$, the following hold.*

(i) *For $1 \leq i \leq m$,*

$$\mathbb{P}(S^i_n \geq c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha \qquad \text{and} \qquad \mathbb{P}(S^i_n \leq -c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha .$$

(ii) *If $|\langle S^i, S^j\rangle_n| \leq \delta$ on Ω for all $i \neq j$, and $\beta^2\delta + \beta^3 b \leq 1$, then*

$$\mathbb{P}\Big(\max_{1\leq i\leq m} S^i_n < c\beta\Big) \leq C\big(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha}\big) .$$

Proof. Let $(\mathcal{G}_t)_{0\leq t\leq n}$ be the filtration, let τ be the least integer $0 \leq t < n$ such that $\langle S^i\rangle_{t+1} > 2C_0$ for some i, and let $\tau = n$ if there is none. Since brackets are predictable, τ is a stopping time. On Ω we have $\tau = n$, so replacing each S^i by S^i stopped at τ changes neither the hypotheses nor S^i_n on Ω, and gives $\langle S^i\rangle_n \leq 2C_0$. Let $V_{ij} = \langle S^i, S^j\rangle_n$, so that $|V_{ij}| \leq 2C_0$ by the Cauchy–Schwarz inequality. We take $c \leq \min\{1/4, c_0/8\}$ small and C large, both depending only on c_0 and C_0.

Let S be a martingale with respect to $(\mathcal{G}_t)$, with $S_0 = 0$, increments at most b_1 and $\langle S\rangle_n \leq 8C_0$, let $|s|b_1 \leq 1/2$, and let

$$\psi(s) := \sum_{t<n} \log\mathbb{E}\big(e^{s(S_{t+1}-S_t)} \bigm| \mathcal{G}_t\big) \qquad \text{and} \qquad W(s) := \exp\{sS_n - \psi(s)\} .$$

By successive conditioning $\mathbb{E}W(s) = 1$. Since each increment $S_{t+1} - S_t$ has conditional mean zero and absolute value at most b_1, Taylor's formula for the exponential and the logarithm gives, whenever $|s|b_1 \leq 1$ and with an absolute implicit constant,

$$\log\mathbb{E}\big(e^{s(S_{t+1}-S_t)} \bigm| \mathcal{G}_t\big) = \frac{s^2}{2}\mathbb{E}\big((S_{t+1} - S_t)^2 \bigm| \mathcal{G}_t\big)\big(1 + O(|s|b_1)\big) . \tag{9.2}$$

Summing over $t < n$ gives $\psi(s) = s^2\langle S\rangle_n/2 + O(|s|^3 b_1)$, and the same holds at $2s$. Hence $\psi(2s) - 2\psi(s) \le Cs^2$, and since $W(s)^2 = W(2s)\exp\{\psi(2s) - 2\psi(s)\}$, we get $\mathbb{E}W(s)^2 \le e^{Cs^2}$. We apply these definitions and bounds with $s = \beta$ to S^i, with $b_1 = b$, and to $S^i + S^j$ for $i \neq j$, whose bracket at time n is $V_{ii} + V_{jj} + 2V_{ij} \le 8C_0$, with $b_1 = 2b$; this is possible because $\beta b \le 1/4$. Let ψ_i, W_i and ψ_{ij}, W_{ij} be the corresponding values of $\psi(\beta)$ and $W(\beta)$. Then

$$\psi_i = \frac{\beta^2}{2} V_{ii} + O(\beta^3 b) \qquad \text{and} \qquad \psi_{ij} - \psi_i - \psi_j = \beta^2 V_{ij} + O(\beta^3 b)\,, \tag{9.3}$$

and $\mathbb{E}W_i = \mathbb{E}W_{ij} = 1$, $\mathbb{E}W_i^2 \le e^{C\beta^2}$ and $\mathbb{E}W_{ij}^2 \le e^{C\beta^2}$.

We prove (i). If c is small enough, then on Ω the bound $V_{ii} \ge c_0$ gives $\psi_i \ge c_0\beta^2/4$. If also C is large enough, then on Ω the inequality $W_i \ge 1/2$ implies $\beta S_n^i \ge c_0\beta^2/4 - \log 2$, hence $S_n^i \ge c_0\beta/8 \ge c\beta$. By the Paley–Zygmund inequality, $\mathbb{P}(W_i \ge 1/2) \ge (\mathbb{E}W_i)^2/(4\mathbb{E}W_i^2) \ge e^{-C\beta^2}/4$, so $\mathbb{P}(S_n^i \ge c\beta) \ge \mathbb{P}(\Omega \cap \{W_i \ge 1/2\}) \ge e^{-C\beta^2}/4 - \alpha$. The same argument applied to $-S^i$ gives the second bound in (i).

We prove (ii). Let $i \neq j$. The exponent in $W_iW_j = W_{ij}\exp\{\psi_{ij} - \psi_i - \psi_j\}$ is at most $\beta^2\delta + C\beta^3 b$ on Ω, and at most $C\beta^2$ everywhere, because $|V_{ij}| \le 2C_0$ and $\beta b \le 1$. So the Cauchy–Schwarz inequality on Ω^c and $\beta^2\delta + \beta^3 b \le 1$ give

$$\mathbb{E}(W_iW_j) \le e^{\beta^2\delta + C\beta^3 b} + e^{C\beta^2}\bigl(\mathbb{E}W_{ij}^2\bigr)^{1/2}\alpha^{1/2} \le 1 + C(\beta^2\delta + \beta^3 b) + e^{C\beta^2}\sqrt{\alpha}\,.$$

Let $\overline{W} := m^{-1}\sum_{i=1}^m W_i$. Then $\mathbb{E}\overline{W} = 1$, and the bounds on $\mathbb{E}W_i^2$ and on $\mathbb{E}(W_iW_j)$ give

$$\operatorname{Var}(\overline{W}) = \frac{1}{m^2}\sum_{i,j=1}^m \bigl(\mathbb{E}(W_iW_j) - 1\bigr) \le C\bigl(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha}\bigr)\,.$$

On Ω, if $\overline{W} \ge 1/2$, then $W_i \ge 1/2$ for some i, and hence $\max_i S_n^i \ge c\beta$, as in the proof of (i). So Chebyshev's inequality gives $\mathbb{P}(\max_i S_n^i < c\beta) \le 4\operatorname{Var}(\overline{W}) + \alpha$, which proves (ii), because $\alpha \le \sqrt{\alpha}$. ☐

9.1. **Lower bounds on the radii and their difference in the plane.** Up to additive constants, $R_{\text{in}}(n) \le R_{\text{mom}}(n) \le R_{\text{out}}(n)$, and by (8.1) the difference $R_{\text{mom}}(n) - r_n$ is, to first order, proportional to $\mathcal{Q}_n$. So Theorem 1.3 (i) needs deviations of $\mathcal{Q}_n$ of about $(\log n)^{1/2}$ standard deviations with probability at least n^{-p}. Lemma 9.1 (i) gives them, because on Ω_n the bracket of $\mathcal{Q}$ is close to the deterministic value $(4\pi\kappa/5)r_n^5$. For part (iv) we use the martingale $Z_n = X_n + \kappa\sum_{x\in A_n} u_x$ of (4.1). The vectors u_x cancel over the sites inside the inner disk, so $|Z_n - X_n|$ is at most $\kappa(R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2) + Cr_n$ when both radii lie between $r_n/2$ and $2r_n$. Lemma 9.1 (i) for the first coordinate of Z, whose bracket lies between two constant multiples of n on every path, gives part (iv)(a); the martingale law of the iterated logarithm for the same coordinate gives part (iv)(b), and the martingale central limit theorem for Z gives part (iv)(c).

Proof of Theorem 1.3 (i) *and* (ii). Comparison with unit cells gives $\sum_{|x|<r}|x| = d\omega_d r^{d+1}/(d+1) + O(r^d)$. Since every lattice site in $B(0, R_{\text{in}}(n))$ belongs to A_n and $A_n \subset B(0, R_{\text{out}}(n)+1)$, taking $(d+1)$st roots gives, for every path,

$$R_{\text{in}}(n) \le R_{\text{mom}}(n) + C \qquad \text{and} \qquad R_{\text{out}}(n) \ge R_{\text{mom}}(n) - C\,. \tag{9.4}$$

For $d = 2$, the two signs in (8.3) therefore give (1.4), with constant $\sqrt{2}/\sqrt{20\pi\kappa} = 1/\sqrt{10\pi\kappa}$.

For the polynomial lower bounds, stop $\mathcal{Q}$ when the walk first leaves $B(0, Kr_n)$, with K as in Section 5.1, and denote the result by $\widehat{\mathcal{Q}}$. Its increments are at most $4Kr_n$, and it agrees with $\mathcal{Q}$ through time n on Ω_n. The bracket calculation in Step 1 of the proof of Theorem 8.1, using (9.1) for the error in the profile, gives on Ω_n

$$\left|\langle\widehat{\mathcal{Q}}\rangle_n - \frac{4\pi\kappa}{5} r_n^5\right| \le C\big(r_n^{9/2}(\log n)^{3/2} + r_n^4\big) = o(r_n^5)\,.$$

Thus Lemma 9.1 (i) applies to $\widehat{\mathcal{Q}}/r_n^{5/2}$ with $b = 4Kr_n^{-3/2}$ and $\beta = a_p\sqrt{\log n}$. Choose $a_p > 0$ so small that the constant C in that lemma satisfies $Ca_p^2 \le \min\{p, 10\}/2$. For all large n, each of the events

$$\{\mathcal{Q}_n \ge ca_p r_n^{5/2}\sqrt{\log n}\} \cap \Omega_n \quad \text{and} \quad \{\mathcal{Q}_n \le -ca_p r_n^{5/2}\sqrt{\log n}\} \cap \Omega_n$$

has probability at least $n^{-\min\{p,10\}/2}/4 - O(n^{-10}) \ge n^{-p}$. On Ω_n, $|X_n|^2 = O(r_n^2)$ and $R_{\mathrm{mom}}(n) = O(r_n)$. Consequently (8.1) and

$$R_{\mathrm{mom}}(n)^3 - r_n^3 = \frac{3}{2\pi}\left(\sum_{x\in A_n} |x| - \frac{n}{2\kappa}\right)$$

show that the first event forces $R_{\mathrm{mom}}(n) - r_n \ge c\sqrt{r_n \log n}$, and the second forces $r_n - R_{\mathrm{mom}}(n) \ge c\sqrt{r_n \log n}$. Now (9.4) proves (1.3). ☐

Proof of Theorem 1.3 (iv). Let $d = 2$, $V_n = \sum_{x\in A_n} u_x$, and $Z_n = X_n + \kappa V_n$, the martingale of (4.1). Its increments are bounded by 2, and the conditional second-moment calculation in Step 1 of the proof of Theorem 8.1 gives, for every $q \in \mathbb{R}^2$,

$$\langle q \cdot Z\rangle_n = \frac{|q|^2 n}{2} - \kappa^2 \sum_{x\in A_n} (q \cdot u_x)^2\,. \tag{9.5}$$

For $Y = e_1 \cdot Z$, it follows that $(1/2 - \kappa^2)n \le \langle Y\rangle_n \le n/2$ on every path.

Step 1: Comparison with the radii. Suppose that $R_{\mathrm{in}}(n) \ge r_n/2$ and $R_{\mathrm{out}}(n) \le 2r_n$. The vectors u_x at the lattice sites with $|x| < R_{\mathrm{in}}(n)$ sum to zero by symmetry, and all these sites belong to A_n. Comparison of the cells of the remaining sites of A_n with the annulus $\{R_{\mathrm{in}}(n) - 1 \le |v| \le R_{\mathrm{out}}(n) + 1\}$, using $|u_x - u_v| \le C/r_n$ on each cell, gives for every unit vector e

$$e \cdot V_n \le \int_{R_{\mathrm{in}}(n)-1 \le |v| \le R_{\mathrm{out}}(n)+1} (e \cdot u_v)_+ \,\mathrm{d}v + Cr_n = R_{\mathrm{out}}(n)^2 - R_{\mathrm{in}}(n)^2 + O(r_n)\,.$$

Here the angular integral is $\int_0^{2\pi} (\cos\theta)_+ \,\mathrm{d}\theta = 2$. Taking e in the direction of V_n gives, under the assumption on the radii,

$$|V_n| \le R_{\mathrm{out}}(n)^2 - R_{\mathrm{in}}(n)^2 + Cr_n\,. \tag{9.6}$$

Step 2: The polynomial lower bound. Lemma 9.1 (i) applies to $Y/\sqrt{n}$ on the whole probability space, with $b = 2/\sqrt{n}$ and $\beta = a_p\sqrt{\log n}$. As in the preceding proof, choosing $a_p > 0$ sufficiently small makes the event $\{Y_n \ge ca_p\sqrt{n \log n}\} \cap \Omega_n$ have probability at least n^{-p} for all large n. On Ω_n, for large n, we have $R_{\mathrm{in}}(n) \ge {}^{r_n}\!/_2$ and $R_{\mathrm{out}}(n) \le 2r_n$. Thus, on the event above, $|X_n| = O(r_n)$ and (9.6) give

$$R_{\mathrm{out}}(n) - R_{\mathrm{in}}(n) \ge c\frac{\sqrt{n \log n}}{r_n} - C \ge c\sqrt{r_n \log n}\,,$$

since $n = (4\pi\kappa/3)r_n^3$. This proves (1.6).

Step 3: The iterated logarithm and small differences. Theorem 1.1 gives $|A_n| = O(r_n^2) = o(n)$ and $R_{\mathrm{in}}(n) + R_{\mathrm{out}}(n) \sim 2r_n$ almost surely. Hence (9.5), the bounded increments, Stout's law, and the martingale central limit theorem with the Cramér–Wold device give

$$\limsup_{n\to\infty} \frac{Y_n}{\sqrt{n \log\log n}} = 1 \quad \text{almost surely}, \qquad \frac{Z_n}{\sqrt{n}} \xrightarrow{\mathrm{d}} N \sim \mathcal{N}(0, {}^1\!/_2 I_2)\,.$$

Since $X_n = o(\sqrt{n})$, the first assertion and (9.6) give

$$\limsup_{n\to\infty} \frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n \log\log n}} \geq \frac{1}{2\kappa}\sqrt{\frac{4\pi\kappa}{3}} = \sqrt{\frac{\pi}{3\kappa}}\,.$$

For the small-ball bound, the same comparison gives

$$\frac{|Z_n|}{\sqrt{n}} \leq \sqrt{\frac{3\kappa}{\pi}}\,\frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n}}(1+o(1)) + o(1) \quad \text{almost surely}\,.$$

Since $\mathbb{P}(|N| \leq t) = 1 - e^{-t^2}$, the portmanteau theorem and continuity of this function show that the upper limit of $\mathbb{P}(R_{\text{out}}(n) - R_{\text{in}}(n) \leq a\sqrt{r_n})$ is at most $1 - e^{-3\kappa a^2/\pi}$. ☐

9.2. **Upper bounds on the local times away from the boundary.** At sites y with $|y| \leq \theta r_n$, where $0 < \theta < 1$, the set $D_n \setminus B(0, R_{\text{in}}(n))$ has small potential, because it has small volume and lies at distance of order r_n from y; only the error in (6.2) remains. In the plane the resulting bound improves Theorem 1.2 (iii).

Proposition 9.2. *Let Ψ be the Euclidean norm, let $0 < \kappa < {}^1\!/\!{}_d$, and let $0 < \theta < 1$ and $p > 0$. There exists $C(d, \kappa, \theta, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$\max_{\substack{y\in\mathbb{Z}^d\\ |y|\leq\theta r_n}} \big|\ell_n(y) - 2d\kappa(r_n - |y|)\big| \leq C\begin{cases}\sqrt{r_n}\log n, & d = 2,\\ \sqrt{r_n \log n}, & d \geq 3\end{cases}. \tag{9.7}$$

The same bound holds for all sufficiently large n almost surely.

Proof. Intersect the events of Theorem 1.2 and of Section 5.1, with exponent $\max\{p, 2\}$ in both, and let $E = D_n \setminus B(0, R_{\text{in}}(n))$ as in Section 6. For large n, every y with $|y| \leq \theta r_n$ lies at distance at least $c_\theta r_n$ from E, and Theorem 1.2 (i) and (ii) give $|E| \leq C r_n^d q_n$; hence

$$|U_E(y)| \leq C_\theta |E| r_n^{1-d} \leq C_\theta r_n q_n\,.$$

By Theorem 1.2 (i), $|R_{\text{in}}(n) - r_n| \leq C r_n q_n$, so (1.10) gives, uniformly over $|y| \leq \theta r_n$,

$$\big|U_{D_n}(y) - 2d\kappa(r_n - |y|)\big| \leq C_\theta r_n q_n\,. \tag{9.8}$$

Since $r_n q_n$ is at most the right side of (6.2), the bounds (6.2) and (9.8) prove (9.7) for large n, and the Borel–Cantelli lemma gives the almost-sure statement. ☐

9.3. **Lower bounds on the local times near the origin.** By (3.11) and (9.8), on Ω_n the deviation $\ell_n(y) - 2d\kappa(r_n - |y|)$ at a site y with $|y| \leq {}^{r_n}\!/_2$ equals $-\mathcal{M}_n^y$ up to $O(r_n q_n + \log n)$. So it suffices to find, among about $r_n^{1/8}$ sites at mutual distances at least $r_n^{1/4}$, one at which $-\mathcal{M}_n^y$ exceeds a multiple of $(\log r_n)^{1/2}$ standard deviations. In the plane the martingales at nearby sites are strongly correlated (see Step 3 of the proof of Theorem 8.3), so we use differences of martingales at two sites at distance about $r_n^{1/8}$. The normalized martingales then have brackets bounded above and below and cross brackets of order at most $r_n^{-1/4}$ (Lemma 9.3), and Lemma 9.1 (ii) applies.

Recall the martingales $\mathcal{M}^f$ of Section 3.1 and the function $\Gamma(f_1, f_2)$ of (8.8). Let $\sigma_n := (r_n \log r_n)^{1/2}$ and $k := \lfloor r_n^{1/8} \rfloor$ when $d = 2$, and $\sigma_n := r_n^{1/2}$ when $d \geq 3$. Let $m := \lfloor r_n^{1/8} \rfloor$, and for $1 \leq i \leq m$ let

$$y_i := i\lceil r_n^{1/4}\rceil e_1\,, \qquad f_i := \begin{cases} g_{y_i + ke_1} - g_{y_i}\,, & d = 2,\\ g_{y_i}\,, & d \geq 3\end{cases} \qquad \text{and} \qquad S^i := -\frac{\mathcal{M}^{f_i}}{\sigma_n}\,.$$

Then $|y_i| \leq m\lceil r_n^{1/4}\rceil \leq r_n^{1/2}/2$ for large n.

Lemma 9.3. *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. There exist $0 < c_0(d,\kappa) \le C_0(d,\kappa)$ and $C(d,\kappa) < \infty$ such that, for all sufficiently large n, on Ω_n, for $1 \le i \le m$ and $1 \le j \le m$ with $j \ne i$,*

$$c_0 \le \langle S^i \rangle_n \le C_0 \qquad \textit{and} \qquad |\langle S^i, S^j \rangle_n| \le C r_n^{-1/4} \,.$$

Proof. Write $r = r_n$ and $L = \log r$ when $d = 2$, and $L = 1$ when $d \ge 3$, so $\sigma_n^2 = rL$. We first bound the sums of the absolute values of the covariances and their truncated first moments. In $d \ge 3$, the squared increments of g_{y_i} are bounded by $C(1 + |x - y_i|)^{2-2d}$; their sum is bounded, and their first moment up to distance r is $O(\log r)$ in $d = 3$ and $O(1)$ in $d \ge 4$. In $d = 2$, the increments of f_i are bounded by

$$\begin{cases} C\big((1+|x-y_i|)^{-1} + (1+|x-y_i-ke_1|)^{-1}\big), & |x-y_i| \le 3k, \\ Ck(1+|x-y_i|)^{-2}, & |x-y_i| > 3k, \end{cases}$$

by (3.4) and (3.5). The squares sum to $O(\log r)$, and their sum weighted by $|x - y_i|$ is at most $Ck \log r$. Bounding each product of increments by half the sum of its squares, using $|y_i| \le \sqrt{r}$, and including the finitely supported terms $\Delta f_i \Delta f_j$, we obtain in every dimension

$$\sum_x |\Gamma(f_i, f_j)(x)| \le CL \qquad \text{and} \qquad \sum_x \min\{|x|, r\} |\Gamma(f_i, f_j)(x)| \le C\sqrt{r}\, L \,,$$

where, in $d \ge 3$, the sites with $|x| > r$ contribute at most Cr^{3-d} to the second sum. The same squared-increment bound controls the total first-departure covariance correction by CL. Let

$$\Sigma_{ij} := \frac{2d\kappa}{L} \sum_x \Gamma(f_i, f_j)(x) \,.$$

Replacing $\ell_n(x)$ by $2d\kappa(r - |x|)_+$, with the error (9.1), and then by $2d\kappa r$ gives, uniformly in i, j on Ω_n,

$$\big|\langle S^i, S^j \rangle_n - \Sigma_{ij}\big| \le \frac{C}{rL}\big(L + \sqrt{r}(\log n)^{3/2} L + \sqrt{r}\, L\big) \le C r^{-1/2} (\log n)^{3/2} \,.$$

For $d \ge 3$, Step 2 of the proof of Theorem 8.3 gives $\Sigma_{ij} = 2d\kappa(2G(y_i - y_j) - \mathbf{1}_{\{i=j\}})$. Its diagonal is $2d\kappa(2G(0) - 1) > 0$, and its off-diagonal entries are $O(r^{(2-d)/4})$. In the plane the summation-by-parts identity from that step also applies to the differences f_i, f_j: their values and increments at distance R are $O(k/R)$ and $O(k/R^2)$, so the boundary term tends to zero. Since the supports of Δf_i and Δf_j are disjoint for $i \ne j$, it gives, with $z = y_i - y_j$,

$$\sum_x \Gamma(f_i, f_j)(x) = 2\big(g(z + ke_1) + g(z - ke_1) - 2g(z)\big) - 2\mathbf{1}_{\{i=j\}} \,.$$

Consequently $\Sigma_{ii} = 4\kappa(4g(ke_1) - 2)/\log r \to 4\kappa/\pi$, and for $i \ne j$ the second difference is $O(k^2/|z|^2)$ by (3.4). In every dimension the diagonal entries are bounded above and below by positive constants, while

$$|\Sigma_{ij}| \le C r_n^{-1/4} \qquad (i \ne j) \,. \tag{9.9}$$

Since $r_n^{-1/2}(\log n)^{3/2} = o(r_n^{-1/4})$, the bracket estimate proves the lemma. ☐

Proof of Theorem 1.3 (iii). By (3.5), the increments of S^i are at most $b = Cr_n^{-1/2}$. We apply Lemma 9.1 (ii) with $\Omega = \Omega_n$, the brackets of Lemma 9.3, $\delta = Cr_n^{-1/4}$, $\alpha = O(n^{-10})$, and $\beta = a\sqrt{\log r_n}$, where $a > 0$ is chosen so small that $Ca^2 \le 1/16$ for the constant in Lemma 9.1. Its hypotheses hold for all large n, and $m \asymp r_n^{1/8}$ gives

$$\mathbb{P}\left(\max_i S_n^i < ca\sqrt{\log r_n}\right) \le C\big(r_n^{-1/16} + r_n^{-1/4}\log r_n + r_n^{-1/2}(\log r_n)^{3/2} + r_n^{1/16} n^{-5}\big) \le C r_n^{-1/16} \,.$$

On Ω_n, by (3.11) and (9.8) with $\theta = 1/2$, the deviation $\ell_n(y) - 2d\kappa(r_n - |y|)$ equals $-\mathcal{M}_n^y + O(r_n q_n + \log n)$, uniformly over $|y| \le r_n/2$, and the error is $o(\sigma_n\sqrt{\log r_n})$. Suppose that $S_n^i \ge ca\sqrt{\log r_n}$.

If $d \geq 3$, it follows that the deviation at y_i is at least $c\sqrt{r_n \log n}$. If $d = 2$, then $S_n^i = (\mathcal{M}_n^{y_i} - \mathcal{M}_n^{y_i+ke_1})/\sigma_n$, so the deviation at $y_i + ke_1$ exceeds the deviation at y_i by at least $c\sqrt{r_n}\log n$, and one of the two deviations is at least $(c/2)\sqrt{r_n}\log n$ in absolute value. All sites used here lie in $B(0, \sqrt{r_n})$. The failure probability is at most $Cr_n^{-1/16} + O(n^{-10}) \leq n^{-c}$ for some $c > 0$ and all large n, proving (1.5). ☐

9.4. **Logarithmic lower bounds for the radii.** If D_n contains $B(0, r_n - h)$, then the walk has visited every site of an annulus of width of order h inside the sphere of radius r_n. If $R_{\text{out}}(n) \leq r_n + h$, then the martingale $\mathcal{Q}$ of (8.1) shows that, except with probability at most $\exp\{-cr_n^{d-3}h^2\}$, the walk has visited at least $cr_n^{d-1}h$ sites of an annulus of width of order h near this sphere; this probability is small when $d \geq 3$ and h has order $\log n$. From each such site the walk leaves $B(0, r_n + h)$ along a straight path with conditional probability at least e^{-Ch}, so the probability that it follows none of these paths from cr_n^{d-1} first visits on disjoint time intervals is at most $\exp\{-cr_n^{d-1}e^{-Ch}\}$.

Proposition 9.4 (Lower bounds for the radii)**.** *Let Ψ be the Euclidean norm, and let $0 < \kappa < 1/d$. There exist $c(d, \kappa) > 0$, $C(d, \kappa) < \infty$ and $h_0(d, \kappa) > 0$ such that the following hold.*

(i) Tail bounds*: for all sufficiently large n and every real h with $h_0 \leq h \leq r_n/8$,*

$$\mathbb{P}\big(R_{\text{in}}(n) \geq r_n - h \text{ and } R_{\text{out}}(n) \leq r_n + h\big) \leq \exp\big\{-cr_n^{d-1}e^{-Ch}\big\}, \tag{9.10}$$

$$\mathbb{P}\big(R_{\text{out}}(n) \leq r_n + h\big) \leq \exp\big\{-cr_n^{d-1}e^{-Ch}\big\} + \exp\big\{-cr_n^{d-3}h^2\big\}. \tag{9.11}$$

(ii) Almost-sure bounds*: almost surely, for all sufficiently large n,*

$$\max\{r_n - R_{\text{in}}(n), R_{\text{out}}(n) - r_n\} > c\log n, \tag{9.12}$$

and, if $d \geq 3$, also $R_{\text{out}}(n) - r_n > c\log n$.

In the plane the last term in (9.11) is not small when h has order $\log n$.

Proof. Step 1: Trials. Every conditional transition probability is at least $p_0 = 1/(2d) - \kappa/2 > 0$. Put $T = \lceil 4h\sqrt{d}\rceil$. From any site x with $|x| > r_n - 2h$, taking T steps in the outward direction of a coordinate of largest absolute value leaves $B(0, r_n + h)$: along this path the distance from the origin increases at rate at least $d^{-1/2}$. The conditional probability of following the path is at least p_0^T.

Fix a deterministic $\Lambda \subset \{x \in \mathbb{Z}^d : |x| > r_n - 2h\}$. Start a trial at each first visit to a site of Λ, discarding the first visits after time $n - T$ and those before the preceding trial ends. If $|A_n \cap \Lambda| \geq k > T$, at most T first visits are too late, and each trial interval contains at most $T + 1$ first visits, so at least $j = \lfloor (k - T)/(T + 1)\rfloor$ trials start. Successive conditioning at their starting times bounds the probability that the first j trials all fail by $(1 - p_0^T)^j$. A successful trial leaves $B(0, r_n + h)$ by time n, so

$$\mathbb{P}\big(R_{\text{out}}(n) \leq r_n + h \text{ and } |A_n \cap \Lambda| \geq k\big) \leq (1 - p_0^T)^{\lfloor (k-T)/(T+1)\rfloor}. \tag{9.13}$$

Step 2: The tail bounds. After increasing h_0, the set $\{x \in \mathbb{Z}^d : r_n - 2h < |x| < r_n - h - 1\}$ contains at least $cr_n^{d-1}h$ sites, all of which belong to A_n when $R_{\text{in}}(n) \geq r_n - h$; so (9.13) for this set proves (9.10). For $\Lambda \coloneqq \{x \in \mathbb{Z}^d : r_n - 2h < |x| \leq r_n + h\}$, (9.13) gives, for every fixed $c_1 > 0$,

$$\mathbb{P}\big(R_{\text{out}}(n) \leq r_n + h \text{ and } |A_n \cap \Lambda| \geq c_1 r_n^{d-1}h\big) \leq \exp\big\{-cr_n^{d-1}e^{-Ch}\big\}, \tag{9.14}$$

where c can depend on c_1.

It remains to bound the probability that $R_{\text{out}}(n) \leq r_n + h$ and $|A_n \cap \Lambda| < c_1 r_n^{d-1}h$. Jensen's inequality gives $|x| \leq \int_{C_x} |v|\,\mathrm{d}v$. For $h_0 \geq \sqrt{d}$, the cells of the sites with $|x| \leq r_n - 2h$ lie in $B(0, r_n - h)$, so these sites contribute at most

$$\int_{B(0,r_n-h)} |v|\,\mathrm{d}v \leq \int_{B(0,r_n)} |v|\,\mathrm{d}v - c_2 r_n^d h = \frac{n}{2\kappa} - c_2 r_n^d h$$

to $\sum_{x\in A_n}|x|$. On $\{R_{\text{out}}(n)\le r_n+h\}$ the other sites contribute at most $2r_n|A_n\cap\Lambda|$. Take $c_1=c_2/4$. Then (8.1) shows that, on the event that $R_{\text{out}}(n)\le r_n+h$ and $|A_n\cap\Lambda|<c_1r_n^{d-1}h$,

$$\mathcal{Q}_n\le -\kappa c_2r_n^dh+(r_n+h)^2\le -cr_n^dh$$

for large n, after increasing h_0 if $d=2$. Stop $\mathcal{Q}$ at the first time t with $|X_t|>r_n+h$; its increments are bounded by $4(r_n+h)\le 5r_n$, and it agrees with $\mathcal{Q}$ through time n on this event. Azuma's inequality bounds the probability of this event by $\exp\{-cr_n^{2d}h^2/(nr_n^2)\}\le\exp\{-cr_n^{d-3}h^2\}$. Together with (9.14), this proves (9.11).

Step 3: The almost-sure bounds. Take $h=c_3\log n$ with $c_3>0$ small enough. The first tail is then summable in every dimension, and the second is summable for $d\ge 3$. The Borel–Cantelli lemma proves (ii), after decreasing c to at most c_3. ☐

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